

Engineers guide to Terfenol-D actuators: Design, performance, and real-world applications

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ABSTRACT

Terfenol-D, a rare earth substance renowned for its remarkable magnetostrictive capabilities, is used in the field of actuators and sensors. The objective of this study is to demonstrate the most recent developments in Terfenol-D actuator technology and its use in many fields such as motors, fuel injectors, inkjet printing heads, servovalves, pumps, and active vibration controllers. The emphasis will be on showcasing the latest accomplishments in this domain. This study offers comprehensive insights into the design, operational features, and performance metrics of several Terfenol-D actuators, accompanied by pertinent schematic illustrations and quantitative measurements. This paper provides an overview of the fundamental structures, consistency of the magnetic field along the Terfenol-D rod, displacement amplification process, and several applications. The focus of the argument has been on the ongoing scientific studies about the actuation capabilities of Terfenol-D actuators. This review paper aims to appeal to the interests and passion of academics, researchers, and engineers involved in the manufacturing, design, analysis, and control of Terfenol-D actuators.

1. Introduction

Each word in Terfenol-D has a specific significance. For instance, “Ter” represents the element terbium, “Fe” is the chemical symbol for iron, “Nol” stands for Naval Ordnance Laboratory, and “D” represents dysprosium. Terfenol-D has many notable characteristics, such as a substantial electromechanical coupling coefficient of 0.73, a significant magnetostrictive strain ranging from 800 to 1600 ppm, a rapid reaction time, a high energy density, and a considerable output force. Schematically Fig. 1 shows the meaning of Terfenol-D.

Terfenol-D-based actuators are widely employed in precision positioning by [2–8], in studies of fuel injector by [9–15], active vibration control by [16–34], DoD ink jet printer for droplet formation by [35], and cutting tools for machining and high precision machining by [36–38], research work in motors by [39–45], pump by [46–52], valve by [53–58], brakes by [59].

As a result, in the past 80 years the number of papers in this field of research has increased tremendously. But the quantity of publications has decreased unexpectedly in recent years. The causes for this decrease in publications are a lack of modelling and control of Terfenol-D actuator nonlinear dynamics, nonlinear hysteresis, temperature drift, and

the necessity for a feasible displacement amplification mechanism. Figs. 2 and 3 shows the annual number of publications on Terfenol-D research materials, as reported by the Scopus search engine. In addition, the contribution of various nations to knowledge development in this field of Terfenol-D materials is given in the Fig. 3.

Only a few review articles on Terfenol-D actuation work, such as hysteresis [60], vibration energy harvesters [61], fuel injection actuation [62], structural vibration control [63] and modelling and control [64], as well as a review on the art of characterization, material development, and modelling of iron-gallium alloy [65], are available in the open literature. Nonetheless, there is a scarcity of review publications focusing on various displacement amplification techniques and actuator applications. A comparable report or review is not accessible in open literature to the best of the author’s knowledge. Fig. 4 shows the flow chart for current review work.

The remainder of this work is structured as follows, based on the aforementioned reviews. Section 2 is based on development of Terfenol-D material and alloying of Terfenol-D is given in Section 3. A brief review about the manufacturing of Terfenol-D is written in Section 4. For procurement of Terfenol-D rods, the suppliers is listed in Section 4. The various design consideration which is required to get high displacement

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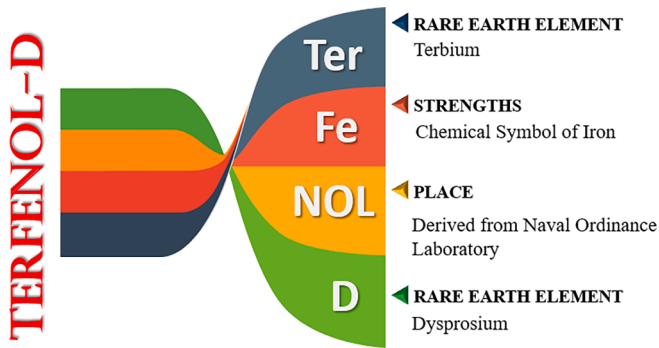


Fig. 1. Meaning of Terfenol-D [1].

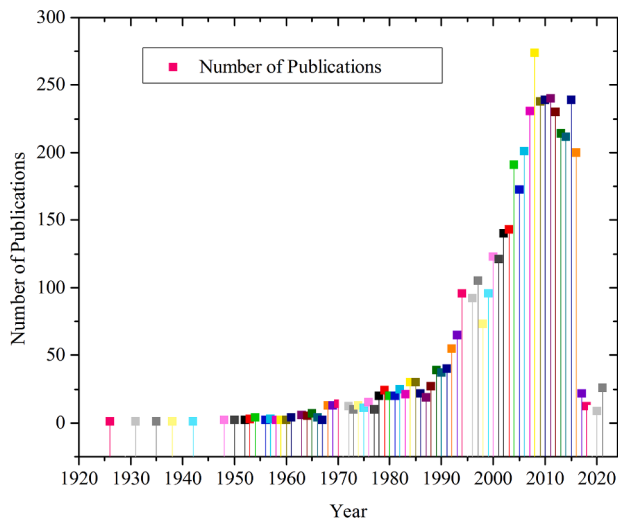


Fig. 2. The number of papers on the subject of magnetostrictive materials published per year, according to Scopus' search engine (as of January 2023).

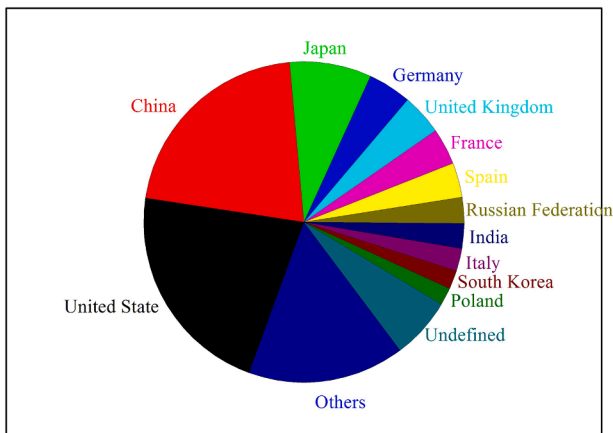


Fig. 3. Country-specific knowledge in the realm of magnetostrictive materials (based on Scopus data from 1926 to 2023).

output such as magnetic field, prestress is discussed in Section 5 and for optimization of these parameters in Section 6. Section 7 will explicate the different displacement amplification mechanisms used by researchers. Section 8 delves into the many applications of Terfenol-D actuators, such as motors, fuel injectors, inkjet printing heads, servovalves, pumps, and active control vibration. Section 9 concludes with some enlightening insights and recommendations.

2. Terfenol-D ($Tb_xDy_{1-x}Fe_2$) material development

Magnetostriction is the reorientation of a ferromagnetic material's magnetic dipole as a result of an external magnetic field. The schematic for magnetostrictive effect is shown in Fig. 5.

The authors have provided an outline of the chronological progression of magnetostrictive materials in this section. Magnetic materials can exhibit both positive and negative magnetostriction. Researchers like Gray [66] and Clark [67] had a noteworthy role in the development of magnetostrictive materials. The pure element nickel has the most negative magnetostriction (-40 ppm). Alfer alloy ($Al_{0.3}Fe_{0.87}$) was created by Japanese scientists as a nickel substitute. Alfer has around 40 ppm of positive magnetostriction. Subsequently, researchers / scientific professionals made efforts to decrease the high temperature magneto-crystalline anisotropy. Fig. 6 shows the brief timeline for magnetostrictive materials.

Cullen, Clark, and his colleagues had previously studied on the magnetostriction of rare earth-iron, RFe_2 compounds at exceptionally high temperatures. To generate substantial magnetostriction at ambient temperature, researchers initially focused on R_2CO_{17} molecules. The highest magnetostriction was found in the Pr_2CO_{17} at room temperature (336 ppm). Then further study focused on the high Curie-temperature RFe_2 metals. Significant magnetocrystalline anisotropy is linked to large magnetostriction. A quest for magnetostrictive materials [68] with high magnetostriction and low anisotropy at room temperature led to the alloying of rare earths with transition metals, culminating in the discovery of enormous room temperature magnetostriction in the Laves phase compound $TbFe_2$ in 1971. $TbFe_2$ had the strongest positive magnetostriction while $SmFe_2$ had the strongest negative magnetostriction at ambient temperature within and between RFe_2 compounds. As terbium's negative magnetocrystalline anisotropy and dysprosium's positive anisotropy compensate for the alloy's anisotropy, terfenol-D anisotropy diminishes. The Terfenol-D alloy was discovered as a result of this. Nominal strains are given in Fig. 7.

Two key explanations for significant magnetostrictions in Terfenol-D are rhombohedral distortions and easy axes in [111] direction. Magnetostriction is reduced as a result of tetragonal distortion in the [100] direction. To obtain maximal magnetostriction in Terfenol-D alloy, researchers put forth a lot of effort to manufacture flawless single crystals. Growing single crystals is an extremely challenging undertaking in practise. The improvement of $Tb_{0.4}Dy_{0.6}Fe_2$ magnetostriction is attributed to the lowering of rhombohedral distortion by replacement of Dy by Tb [69]. Crystals are oriented and solutes or phases migrated using very uniform magnetic fields in order to provide a stronger magnetostrictive behaviour in $(Tb, Dy)Fe_2$ phase, at the lower parts of the alloys [70]. Hari et al. [71] reported a large negative magnetostriction (-1572×10^{-6}) compensation with $Sm_{0.88}Nd_{0.12}Fe_{1.93}$ polycrystalline compound at ambient temperature.

3. Alloying of Terfenol-D

The original Terfenol-D alloy has a serious brittleness flaw. It is so obvious that the researchers employed diverse ways for processing Terfenol-D material, including standard P/M approach, alloys of multiple elements, directed solidification, and single crystal growth. The notable elements utilised as a substitute for Tb, Fe, and Dy include holmium, vanadium, cobalt, manganese, aluminium, niobium, and zirconium. The addition of 15% holmium [72] to Terfenol-D material reduces magnetic anisotropy, hysteresis, saturation in magnetostriction, and the needed driving magnetic field. Shifting the Tb/Dy ratio has expanded the working temperature range of $(Tb, Dy)Fe_2$ alloys. The addition of cobalt or vanadium components has been shown to raise the Curie temperature to around 50°C, the anisotropy compensation temperature, and the material strength.

Aluminum is recognised for its great ductility and resistivity, as well

as its strong saturation magnetostriction, low anisotropy, and big strain coefficient. There is a 50% rise in electrical resistivity, a high piezomagnetic coefficient ($d_{33}=80$ nm/A), magnetostriction up to 800 ppm, and a substantial magnetomechanical coupling when 10% aluminium is added. The anisotropy compensation temperature and the Curie temperature both drop when manganese is added. Because Mn impacts magnetocrystalline anisotropy, alloying Mn for Fe is effective in improving the magnetostrictive behaviour of $\text{Tb}_{0.3}\text{Dy}_{0.7}\text{Fe}_2$. Because of the rise in Curie temperature and decrease in spin reorientation temperature, substituting Co or Mn for Fe can expand the operational temperature range to $\text{Tb}_{0.36}\text{Dy}_{0.64}\text{Fe}_2$. In a 3 kOe magnetic field under 10 MPa pre-stress at ambient temperature, and throughout a wide temperature range from -80°C to 100°C , giant magnetostriction more than 1450 ppm [73] was achieved in the orientated $\text{Tb}_{0.36}\text{Dy}_{0.64}(\text{Fe}_{0.70}\text{Co}_{0.30})_2$ rod. Due to the creation of the TiFe_2 paramagnetic phase, magnetostriction in Terfenol-D material has enhanced with titanium associability ($x = 0.025$) [74].

Fig. 8 shows the the summary of the alloying element to Terfenol-D material. The anisotropy compensated pseudo-binary C15 type Laves phase $(\text{Tb}, \text{Dy})\text{Fe}_2$ is formed through a peritectic process. In addition, $(\text{Tb}, \text{Dy})\text{Fe}_2$ has a negative impact on the magnetostrictive characteristics of the alloys. As a result, niobium and zirconium ($x = 0.025$) [75] have been added to the parent alloy to enhance its magnetostrictive characteristics, which is mostly due to a decrease in the volume fraction of the detrimental $(\text{Tb}, \text{Dy})\text{Fe}_3$ phase. The most important limiting issue for Terfenol-D alloy use on a broad scale is its mechanical qualities. The fracture toughness of the alloy increased 1.5-5 times when Nb was added to it, compared to the Nb-free alloy [70]. This is owing to the NbFe_2 phase's development and the RE-rich phase's capacity to suppress fracture progression. Jiang et al. [76] used an annealing treatment on a TbDyFe alloy to relieve the internal stress at the interface between the net-shaped rare earth phase and the RFe_2 phase matrix, resulting in improved magnetostriction. Researchers/academicians [77–79] have investigated fracture toughness, alleviating internal stresses, and the effect of oleic acid on grain refinement, oxidation prevention, and coercivity enhancement in Terfenol-D alloy.

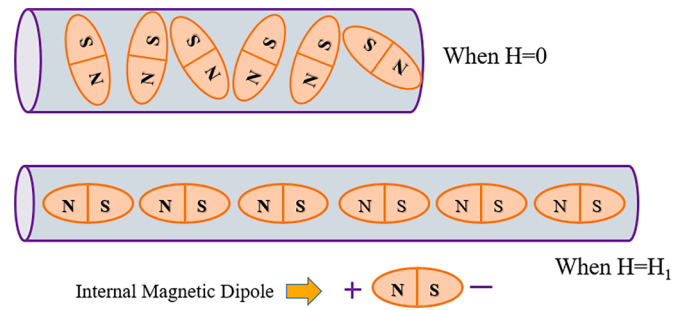


Fig. 5. Schematic for magnetostriction mechanism.

4. Giant magnetostrictive materials (GMMs): from production to market

Free stand zone melting (FSZM), modified Bridgman (MB), sintered powder compact, and polymer matrix composites of Terfenol-D powder procedures are only a few of the Terfenol-D alloy production techniques accessible. However, the ability of free stand zone melting and modified Bridgman to produce Terfenol-D rods with strong magnetostrictions and energy densities is superior. The FSZM and MB processes, on the other hand, are obligated to produce rods with a minimum diameter of 8 and 10 mm, respectively. Powder metallurgical processes offer the potential to produce complicated geometries, particularly small-scale components. The influence of deposition rate and vacuum on thin film creation using the flash evaporation technique was investigated by researchers [80]. The authors [81] designed and implemented the ET-Ryma crystal growth (ECG) system, a next-generation version of the MB method, in 1997. In 1997, a novel electron beam zoning technology [82] was invented, with benefits including a narrow melting zone, low power consumption, and a large temperature gradient. The goal of this procedure is to create a defect-free crystal with a crystallographic direction of $\langle 111 \rangle$ along the driving axis. With ultra high temperature gradient directed solidification apparatus, the author [83] devised a methodology for melting and casting rare earth giant magnetostrictive bars. Because of the persistence of twofold peritectic transition and quick oxidation, obtaining homogeneous morphology in Tb-Fe alloy is extremely challenging. As a result, the author developed a drop tube

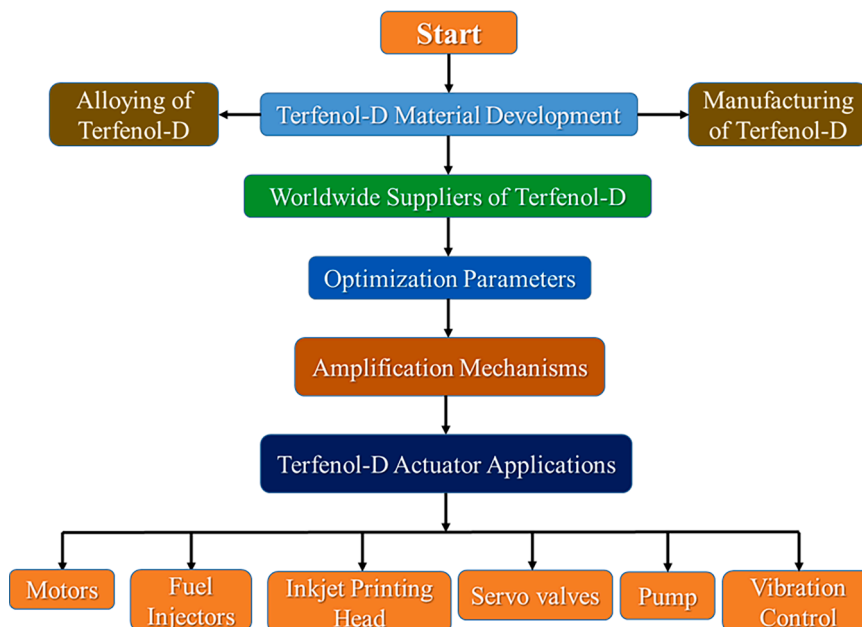


Fig. 4. Flow chart for the current research work.

Year	Timeline for Magnetostrictive Materials
1842	Magnetostriction discovered in Nickel by Joule.
1865	Villari discovered inverse Joule effect.
1926	Anisotropy in single crystal iron.
1965	Rare earth (Terbium & Dysprosium) magnetostriction by Clark.
1972	$TbFe_2$ and $DyFe_2$ at 300^0 K by Clark.
1975	Discovery of Terfenol-D by Clark.
1994	Polymer matrix and Terfenol-D Particulate composite.
1998	Discovery of Galfenol
2002	Oriented particulate composite

Fig. 6. Brief timeline for magnetostrictive materials.

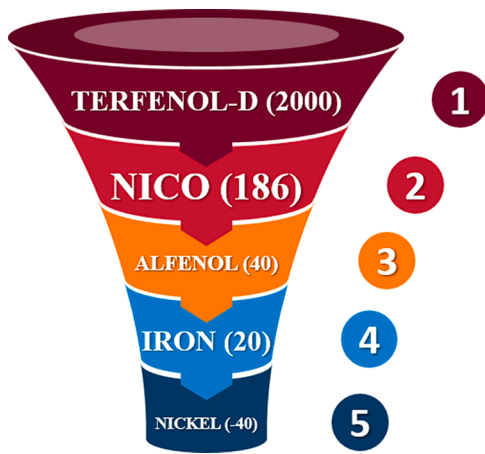


Fig. 7. Magnetostriction values for different materials.

mechanism to achieve homogeneous morphology in Tb-Fe alloy. The drop tube mechanism is detailed in depth someplace [84].

TdVib LLC is one of the most well-known and long-standing providers of large magnetostrictive materials (Old Company is Etrema). In 2016, ETREMA Products, Inc. received complete ownership of the Terfenol-D manufacturing technology from the DOE's Ames Laboratory. Nowadays, more vendors are mostly from China such as Lanzhou Jianxing Commercial and Trading Co., Ltd. China, Suzhou A-one Special Alloy Co., Ltd. China, Tianjin Huaanxuyang Steel Trade Co., Ltd. Beichenqu, Tianjin, China. In the case of magnetostrictive materials, the limited global supplier base is a well-established fact.

5. Magnetic bias and prestress for Terfenol-D actuator

Terfenol-D magnetostriction modulates with mechanical compressive stress, magnetic field strength, and temperature. The Terfenol-D actuator is comprised of three components: the magnetic field generator, prestress mechanisms, and a cooling device. The first step in designing the Terfenol-D actuator is determining the Terfenol-D material dimensions required for output strain and force. The needed length of the Terfenol-D material is commensurate with the output strain and cross-sections area [85–87].

The quadratic strain of Terfenol-D is a distinguishing feature. As a result, the bias magnetic field causes Terfenol-D to partly expand. The sign of the applied longitudinal magnetic field has no effect on the strain of the magnetostrictive material. The Terfenol-D actuator will

experience a strain twice as often as the input current frequency due to the alternating sign input current. By providing a bias magnetization, a linear relationship between the applied coil current and the strain may be established. A permanent magnet or an additional coil can be used to provide this bias magnetization. The Terfenol-D actuator is used in the permanent magnet bias zone where the slope of the curve is greatest [56]. The present authors are investigating another potential of bias magnetization. We supply DC for one coil and AC for another, the magnetic fields of which are integrated to achieve partial magnetization. A cylindrical permanent magnet bias coil configuration was used to supply magnetic bias for the linear magnetostrictive motor [88]. Permanent magnets produce both the driving and bias magnetic fields [89]. Xue et al. [14,90,91], on the other hand, claim that the Terfenol-D actuator's maximum displacement output was attained without the application of a bias magnetic field. As illustrated by several authors [16,56,85–87], there exist several magnetic circuit designs for driving an actuator by a magnetic field.

As a result, low frequency current has been passed through DC and AC driving coils in this configuration [92]. This is due to the formation of eddy current in the conductor, which opposes the AC applied field. As a result, high-frequency magnetic flux will be unable to permeate the material. A bias magnetic field is created using coil 1. Coil 2's magnetic field strength has been overlaid over coil 1's magnetic field. Furthermore, the magnetic coil outperforms the permanent magnet in terms of altering the bias to the required value. We can only obtain a consistent magnetic bias by employing a permanent magnet [93]. Experiments have also shown that superimposing the displacement caused by two coils [94] can result in a substantial displacement from Terfenol-D. AC operation may be pushed to steeper parts of the quasi static curve, where substantially greater strain is realised for a given AC drive level [95], by introducing a DC magnetic bias field. The low reluctance of magnetic flux path value in the C-T-C configuration was obtained by using a high permeability material. As a result, magnetostatic study of magnetostrictive actuators using various housing materials, including aluminium, cast iron, and mild steel, was undertaken [92]. The permeability of actuator housing material affects the magnetic field distribution on Terfenol-D. Because of the poor relative permeability of Terfenol-D material, there is a leakage of static and dynamic flux [39].

The exciting coil current created a positive and negative fluctuating magnetic field, which was countered by the permanent magnet's magnetic field [96,97]. The linear working zone of the GMMs is enlarged by this offset magnetic field. Permanent magnets for bias magnetic fields were utilised in order to avoid multifrequency [17]. Cylindrical permanent magnets [98] have been constructed in order to generate a consistent longitudinal field by modifying the shape or remanence of magnetic rings in each segment. A three-ring permanent magnet [99]

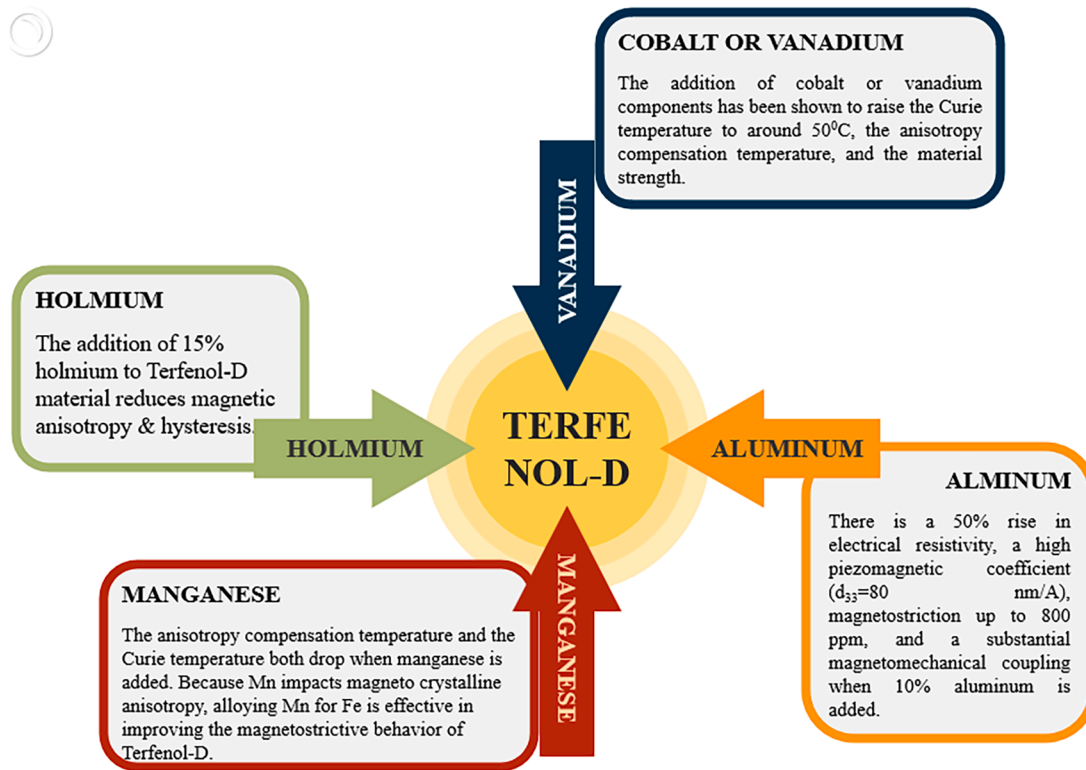


Fig. 8. Alloying of different materials to Terfenol-D.

has been proposed to offer a bias magnetic field and a configuration with high length-to-diameter ratios ($\alpha \leq 20$) in GMMs. It is important to remove the heat generated as a result of the constant activation of the driving coil and the thermal expansion of the Terfenol-D material. As a result, a low-power drive has been suggested and developed [100]. The power saving drive can regulate an intermediate expansion of the Terfenol-D actuator by magnetising and demagnetizing a permanent magnet in a small loop of the B-H curve. In comparison to Nd-Fe-B or Sm-Co permanent magnets, Al-Ni-Co permanent magnets have a higher remanent magnetic flux density, a lower coercive force, and are easier to demagnetize.

The solenoid is the same length as or longer than the GMM rod, resulting in a uniform magnetic field around the rod [101]. A magnetic bias is given by encircling a cylindrical magnet, or a biasing DC current might be given to the solenoid to provide bidirectional strains [86,95]. However, utilising a permanent magnet to apply magnetic bias is successful. Permanent magnets should be utilised for magnetic bias at low

frequencies and in applications requiring relatively high forces and low strains [102].

Magnetostrictive materials are notoriously fragile. Their yield stress when stretched is significantly less than their yield stress when compressed [103]. As a result, the Terfenol-D rod is essentially always subjected to compressive prestress. Mechanical prestress in conjunction with magnetic bias is one of the major variables that improves the total strain capability of Terfenol-D material and generates output in dynamic conditions [95]. Fig. 9 shows the arrangement of prestress for Terfenol-D actuator with plunger and belleville springs. When prestress or axial pressure is applied on the Terfenol-D material, the internal magnetic domains undertaking zero magnetic field arranges along with perpendicular direction of axis of the rod. Following that, an external magnetic field is supplied with prestress, causing most of the magnetic domains to abruptly shift in the direction of the magnetic field, resulting in a higher magnetostriction coefficient and increasing the actuator's output displacement. The schematic for this effect of compressive prestress on

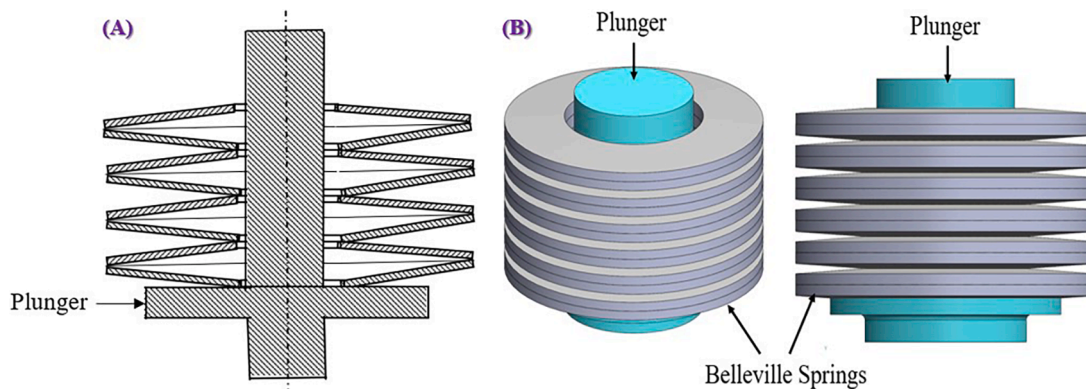


Fig. 9. Model for prestress arrangement with belleville springs. (A) Sectional drawing of series arrangement of belleville springs. (B) 3D drawing of series arrangement of belleville springs.

magnetostrictive effect is shown in Fig. 10.

Terfenol-D is a brittle material. Terfenol-D can sustain pressures of up to 700 MPa, but its tensile strength is only around 28 MPa, which is readily broken, thus it should not be used in tensile or shear stress. In general, disc or Belleville springs excel in mechanical prestressing of Terfenol-D actuators due to their high stiffness, compact volume, and high strain energy density. Also, the spring stiffness should not be too high to absorb the deformation of the Terfenol-D rod. Disc or Belleville springs are frequently stacked in order to improve performance. The load bearing qualities of the springs are improved by stacking them in parallel. Greater deflection is accomplished by stacking them in series. Hysteresis grows with increased prestress, which is one of the major difficulties with mechanical prestress [86,87,95].

6. Optimization of Terfenol-D actuator

Many studies have focused their efforts on improving the performance of the Terfenol-D actuator, but the design of the driving coil, uniform magnetic field along the Terfenol-D rod, and eddy current losses have received greater attention.

6.1. Driving coils design

The fundamental metric for evaluating the Terfenol-D actuator's performance is its electromagnetic conversion efficiency. As a result, the design of the driving coil is given careful consideration. The least dissipated resistive power is an important consideration in the optimal design of drive coils [86]. However, owing to driving coils of the Terfenol-D actuator, these thermal strains are induced [104].

Fig. 11 shows the design steps for driving magnetic coil for Terfenol-D actuator. One should thus pick for the driving coils to guarantee that the beneficial magnetostrictive strains are big compared to thermally produced strains. Thermal parameters should also be addressed for the Terfenol-D actuator's prolonged operating circumstances. To decrease the thermal expansion of the Terfenol-D rod, a water cooling system has been built for the Terfenol-D jet dispenser [105]. They provided a one-dimensional magnetic model that can be used to forecast the magnetic field inside the magnetostrictive driving rod.

6.2. Eddy current losses

For considerable deformation of Terfenol-D material, the driving coil

should have low impedance and pulse current. This is because the coil generates Joule heat by sending a high-frequency current through it constantly, causing eddy currents [94]. The magnetic field in the Terfenol-D rod is non-uniform due to eddy current and skin effect [106]. When using Terfenol-D rods at high frequencies, the materials must be laminated to reduce eddy current losses. Lamination, on the other hand, would limit material use and stiffness [106]. They described how Terfenol-D rod should be laminated if the non-uniformity error or effective magnetic field strength error exceeded 5%. With a broader hysteresis loop, eddy current losses reduce the Terfenol-D actuator's displacement output. Because the Jiles Atherton model fails to estimate losses outside of Terfenol-D, a new model [107] has been devised. Excess eddy current losses owing to non-uniform magnetic flux distribution must be taken into account when calculating real values of eddy current power losses [108].

6.3. Uniformity in magnetic field

One method for increasing the uniformity of the magnetic flux density is to use a hollow Terfenol-D rod, however it is difficult to produce sufficient magnetic bias and is not suitable for long actuators [86]. For convection cooling, Permalloy rings have been arranged in permanent magnet stacks [2]. Because the Permalloy rings guarantee an appropriate least reluctant path for the magnetic flux produced by the magnetic field generating coil. The ideal arrangement between the bobbin and the thickness of the permanent magnet for uniform distribution of magnetic field along the Terfenol-D rod was determined using finite element analysis [109]. Yang et al. [110] has optimised the nonuniformity of the axial direction magnetic field intensity for an electro-hydrostatic actuator made of Terfenol-D material. Two of the most important characteristics for optimising axial magnetic field distribution are the length of the driving coil and the relative permeability of the magnetizers on the magnetic circuit. Rong et al. [111] has undertaken a similar intriguing work on the dynamic performance of Terfenol-D actuators for aviation applications. This is accomplished by creating a Terfenol-D stack actuator in which short Terfenol-D rods and permanent magnets are alternately placed to produce bias field evenness.

7. Displacement amplification mechanisms

The Terfenol-D actuators are known for their reliability, high

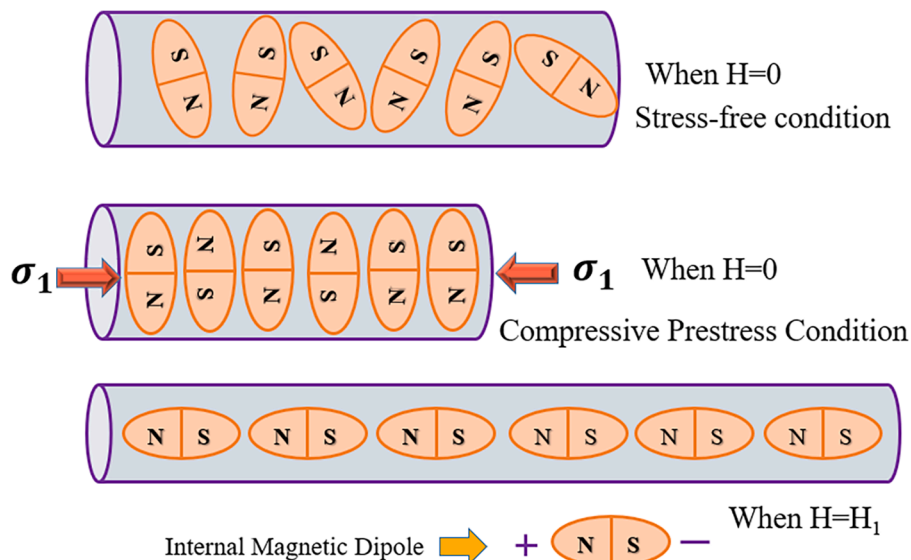


Fig. 10. Influence of prestress on magnetostriction.

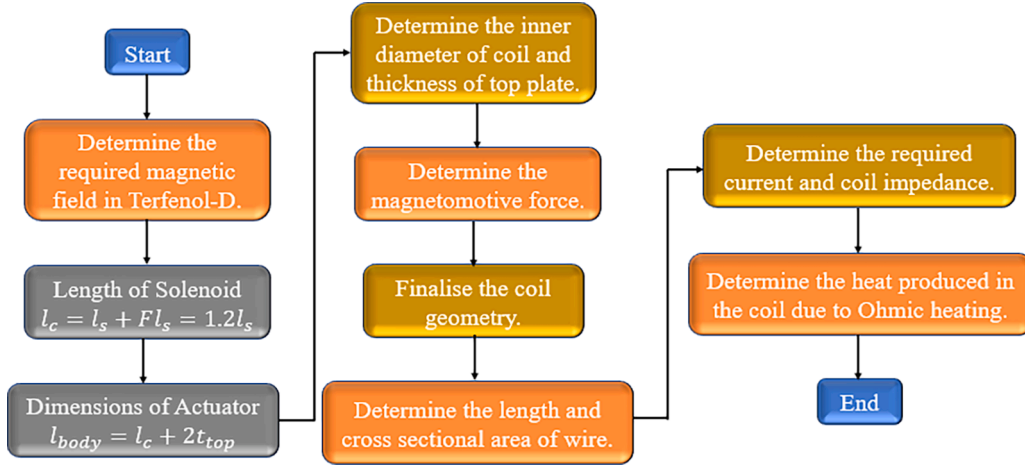


Fig. 11. Flow chart for design steps for driving coil of Terfenol-D actuator.

response time, and high force density. However, a key hurdle is magnifying the relatively modest displacement of Terfenol-D actuator. For displacement amplification, there are a variety of options. There are two types of methods for enhancing the output stroke of magnetostrictive materials: direct and rectification.

Direct amplification techniques amplify the motion created by material activation using a device such as a lever (see Fig. 12). The motion of the driving material and the motion of the output material are synchronous. As an outcome, the mechanism's output vibrations are at the same frequency as the driving material activation. A multistage application of a lever with a shaft pin structure can produce a considerable stroke, however applications are constrained by mechanical clearance and structural division. The Terfenol-D actuator comprises differential lever amplification with synthetic drive [112]. The elastic beam amplification mechanism is small, compact and easy to maintain. It's been utilised in ON/OFF valves in the previously. This construction, on the other hand, may create stress concentration or beam flexing [57], both of which are unfavourable for significant displacement output. As a result, Yang et al. [113,114] developed an innovative hydraulic amplifier (see Fig. 13) that uses a flexible piston to prevent leakage and friction, as well as a gradient connection chamber to reduce pressure loss. The wide film bends horizontally in this amplifier due to the axial movement of the plunger on the Terfenol-D rod. This deformation produces a decrease in volume in the sealed chamber, which causes a rise in pressure, compressing the tiny film and causing a similar deformation. GMM's elongation is eventually increased in this method. However, typical displacement amplification devices are not designed with efficient energy transmission in mind. Low friction, minimal clearance, non-lubrication, great displacement output, and mild stress concentration have all been engineered into an elliptical flexure hinge amplification structure [98,115].

Hydraulic amplification is especially appealing as a means of achieving substantial mechanical gains in a small area. Internal friction and compliance in the fluid chamber components, as well as non-linearity in response, can severely reduce mechanical performance [56,116]. A novel hydraulic displacement amplifier [117] based on the area differential principle and Pascal's law is presented to successfully overcome the bottleneck issue of the Terfenol-D actuator's modest output displacement. The input piston in this displacement amplification device is directly connected to the Terfenol-D output tip, and the output piston is connected to the application area. The Terfenol-D actuator's output displacement will be magnified owing to the hydraulic cylinder's differential area, and the amplified displacement will then be transferred to the application area. The amplification ratio for this device is defined by the ratio of two piston diameters (depending on application). As illustrated in Figs. 14-15, a Terfenol-D actuator is carried by a cylindrical metal holder, with one end immobilised by a screw and the other end attached to the input piston. The input and output pistons will then be linked by a hydraulic cylinder. The output piston's other end will be attached to the interface of a certain application area. To increase reaction speed and motion consistency, a tiny spring is placed to the gap.

When Terfenol-D physically drives the input piston, the fluid in the cylinder is compressed swiftly. The created pressure will be evenly distributed to any part of the closed hydraulic cylinder, according to Pascal's law. The output piston will then be driven to provide increased smooth motion. Finally, this fluid motion will be transmitted to the linked interface of a certain application. The following connection may be established using the equivalent volume concept as Eq. (1).

$$\left(\frac{\pi}{4} \times D_{\text{Input piston}}^2 \times \delta_{\text{Input piston}}\right) = \left(\frac{\pi}{4} \times D_{\text{Output piston}}^2 \times \delta_{\text{Output piston}}\right) \quad (1)$$

Where, $D_{\text{Input piston}}$ = Diameter of input piston, $D_{\text{Output piston}}$ =

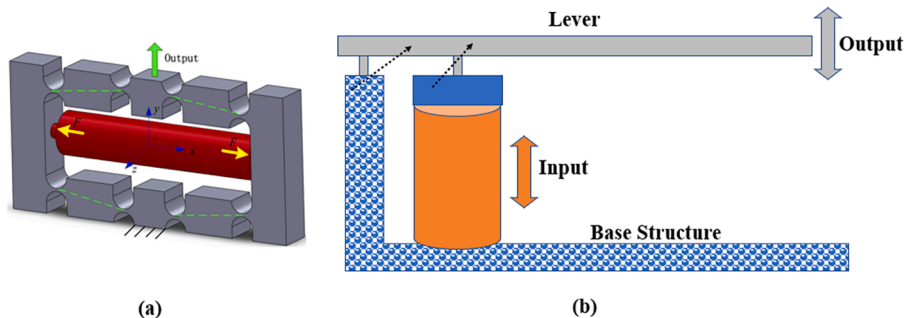


Fig. 12. Topology for direct amplification mechanisms a) Pantograph b) Lever type [112].

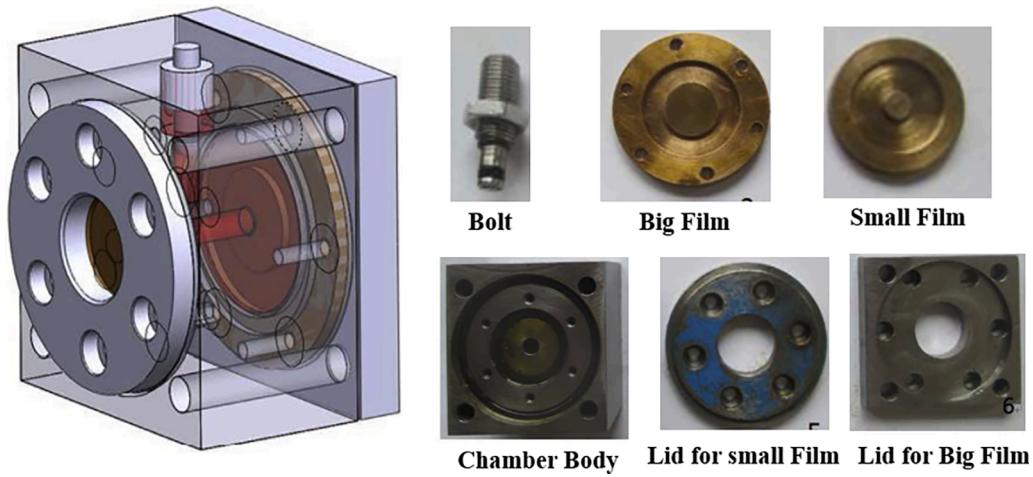


Fig. 13. Schematic for hydraulic displacement amplifier [114].

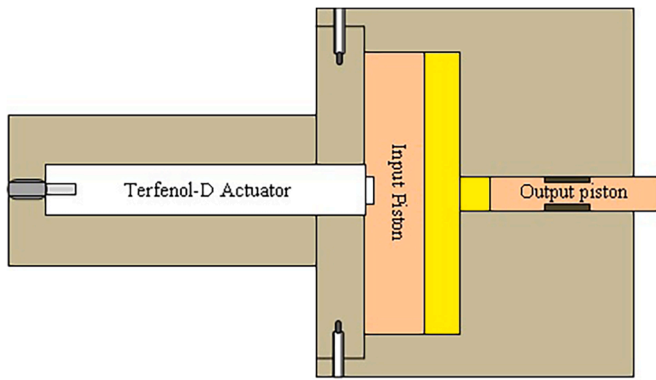


Fig. 14. Schematic for hydraulic displacement amplifier.

diameter of output piston, $\delta_{\text{input piston}}$ = displacement of input piston, $\delta_{\text{Output piston}}$ = displacement of output piston. Then displacement amplification ratio of hydraulic amplifier can be given by Eq. (2).

$$A = \frac{\delta_{\text{Output piston}}}{\delta_{\text{Input piston}}} = \frac{D_{\text{Input piston}}^2}{D_{\text{Output piston}}^2} \quad (2)$$

However, the hydraulic amplification mechanism is bulky and complex in construction, thus a more compact design for aerospace applications has been presented; inventive Lead zirconate titanate (PZT) rings hydraulic amplification technique for piezoelectric actuator. This amplifier is made up of a PZT ring as the active component, which drives the piston hydraulically and generates a high force ($28 \times 10^{-3} \text{ N/V}$) and relatively high displacement ($0.26 \times 10^{-6} \text{ m/V}$). When the voltage is applied, the dimensions of the PZT ring will vary, leading to a change in the internal volume of the actuator being reduced or increased. As a result, the PZT hydraulic actuator may be a viable option to filling the current gaps between the disc stack's high-force-low-displacement arrangement and the low-force-high-displacement bi-morph. For active mounting of the engine [118] a compact bi-directional magnetostrictive actuator with a hydraulic gain was offered. They considered the kinematic gain as one of the essential performance elements of any amplified displacement actuator. The primary drawback of a direct amplification hydraulic version is that the material force decreases the

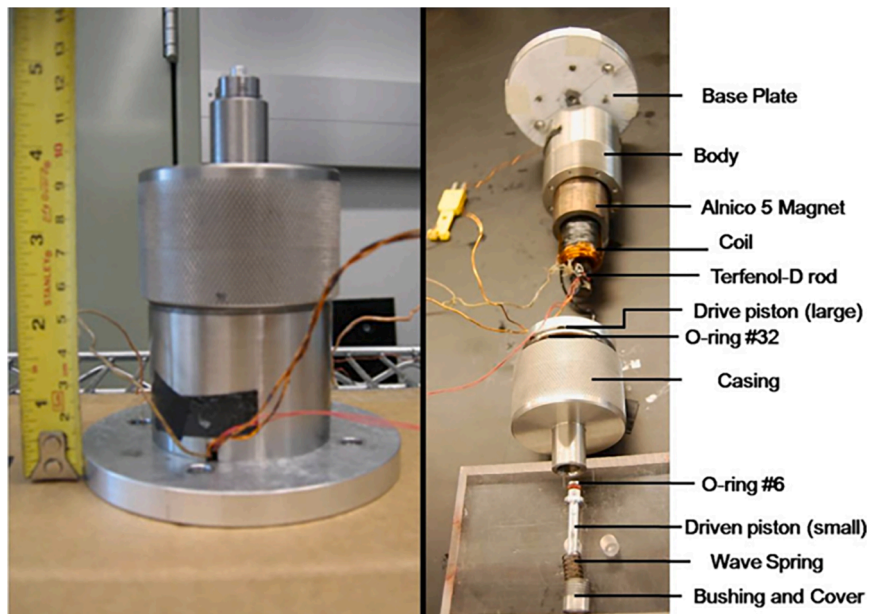


Fig. 15. Assembly of hydraulically amplified Terfenol-D actuator [116].

amplification ratio. A method of rectification [46–49] for displacement amplification has thus been used.

The rectification approach converts the magnetostrictive material's tiny, high frequency vibrations into massive, lower frequency actuator displacements. The resulting displacement is not directly related to the frequency of activated material. These approaches are fundamentally different from direct approaches. Fluid rectification via a one-way control valve converts an intelligent material's high frequency vibration into a hydraulic cylinder's substantial motion. It varies from previous step-and-repeat and inchworm methods of amplification, using friction to capture the material vibrations. Friction-based approaches can have issues with wear or load capacity, particularly when high force output is required. One technique to expand the frequency bandwidth across which the valve can correct the fluid flow is to make the valve smaller. While this is appealing in theory, the solid-fluid hybrid actuation technique suffers from a practical restriction that limits its usefulness. Inertial loads, fluid viscosity, and compressibility, in combination with a loss mechanism inherent in the active material, restrict the driving actuator's effective bandwidth and total output power [18, 119–121]. As a result, Giurgiutiu et al. [119] proposed a generic stiffness idea for making optimal use of an induced strain actuator. When the stiffness ratio matches the critical stiffness ratio, the induced strain actuator's output energy is maximised (optimum stiffness ratio). To obtain the most energy out of a displacement amplified induced strain actuator, you must use its optimal gain, which is provided by Eq. (3),

$$G_{opt} = \frac{1}{\sqrt{r}} \quad (3)$$

Where, r is the ratio of load stiffness to smart material stiffness. Support elasticity can potentially negatively affect the stiffness match concept and reduce output effectiveness [120]. A novel hydraulic actuator design for the Terfenol-D magnetorheological (MR) fluid pump [121] incorporates the tunable rheological behaviour of MR fluid combined with Terfenol-D magnetostrain. The compact actuation mechanism is based on the rectification of resonant vibrations produced by a Terfenol-D pump magnetostrain via MR fluid flow control and the use of hydraulic advantage to convert magnetostrain into significant deflection and force. The use of a driven piston diameter that is larger than the drive piston provides a hydraulic advantage.

As shown in the Table 1, differential and compliant amplification can provide a higher displacement output with a compact structure, although magnification is not optimal. For perfect displacement amplification with high energy output, many researchers have turned to hydraulic amplification techniques. For Terfenol-D pump applications that need a large frequency bandwidth, rectification systems comprising a check valve and a MR valve have been used. Flexible piston hydraulic Terfenol-D mechanisms feature a tiny displacement but a high output force, making them suitable for precise applications. The Terfenol-D electro-hydrostatic actuator, which consists of the electrical-magnetic, magneto-elastic, elastic-hydraulic, and hydraulic-mechanical energy transformation processes, has been manufactured and developed [125]. A rotor valve was used to transform the tiny, high frequency vibration movement of both terphenol-d rods by fluid rectification in a continuous, bidirectional and smooth movement of a hydraulic cylinder [126, 127]. On the basis of inchworm motion, Zhou et al. [128] designed and tested the Terfenol-D rotary actuator for precision machining, demonstrating high precision rotation motion and high speed. Zhou et al. [129] invented and tested the large stroke and high accuracy linear displacement inertial Terfenol-D actuator employing the inchworm motion concept. Afzal et al. [130] suggested the Terfenol-D actuator for the management of normal friction interface load in order to achieve the chosen friction damping in the structure.

8. Different applications of Terfenol-D actuators

With the recognition of Terfenol-D materials' potential for usage in a

Table 1
Comparison of displacement amplification mechanisms.

Reference	Amplification Mechanism	Mechanism Principle	Magnification
Ref. [119]	Hydraulic amplification	Pascal's law	13.35
Ref. [122]	PZT rings hydraulic amplification	Hydraulic	Not reported
Ref. [46, 49, 121]	Electro-hydraulic hybrid actuator	Rectification through Magnetorheological fluid and check valve	Not reported
Ref. [116]	Hydraulic amplification	Pascal's law	Not reported
Ref. [56]	Mechanical amplifier with flexure mechanism	Bridge amplification	Not reported
Ref. [98]	Elliptical Flexure hinge mechanism	Differential amplification	2.91
Ref. [115]	Circular or rectangular hinge mechanism	Differential amplification	10.75
Ref. [114]	Hydraulic amplification through flexible piston	Pascal's law	8.5
Ref. [59]	Mechanical amplification	Secondary lever	5.28
Ref. [123]	Flexure-hinged displacement amplification	Compliant mechanism	8
Ref. [124]	Mechanical amplification	Compliant mechanism	30.5
Ref. [123]	Flexure-hinged displacement amplification	Compliant mechanism	Not reported

variety of applications, greater emphasis has been placed on material development and strain performance enhancement. Various applications have been found, including motors, fuel injectors, active vibration control, brakes, cutting tools, valves, pumps, and so forth.

8.1. Terfenol-D motors

Terfenol-D motors are categorised into two types: inchworm and resonance. Terfenol-D motors provide several benefits over typical electric or electromagnetic motors, including precise control and great torque at low speeds. Static displacement (20 – 200 μm), large forces/torques, and resolution are difficult to achieve with electro-magnetic motors. Because of its capacity to respond very quickly, the Micro actuated Terfenol-D motor has been conceived and developed [109] to produce controlled displacement with nanoscale precision and millimetre structure. This microactuator generates 25 N of force, 3 μm of displacement, 1.5 A of current, and 250 nm of displacement every 0.1 A of current. A stepping motor with a torque of 12.2 Nm, an angular resolution of 800 μrad , and a low output power has been reported [40]. One of the key operating concepts of hybrid linear motors is inchworm motion. Two clamping devices and one pushing actuator are used to achieve worm-like motion. The development of a new hybrid linear motor [39] alleviated the challenges in the clamping and pushing functions. The pushing force is proportional to the cell's friction force. The schematics for the self moving linear motors have been included in Figs. 16–17.

Because it requires a separate active element to clamp, the inchworm approach has high coil losses. Without torque correction [41], it is impossible to achieve tolerance on clamping parts to correct wear. As a result, additional research into the contact mechanism, number of cells, high frequency activation, wear, and temperature impacts is needed in the future for hybrid linear motors. Friction motors deliver motion to a rotor or driven part by using the vibration of a stator. An orthogonal actuator is activated with sinusoidal 900 phase shift currents to generate

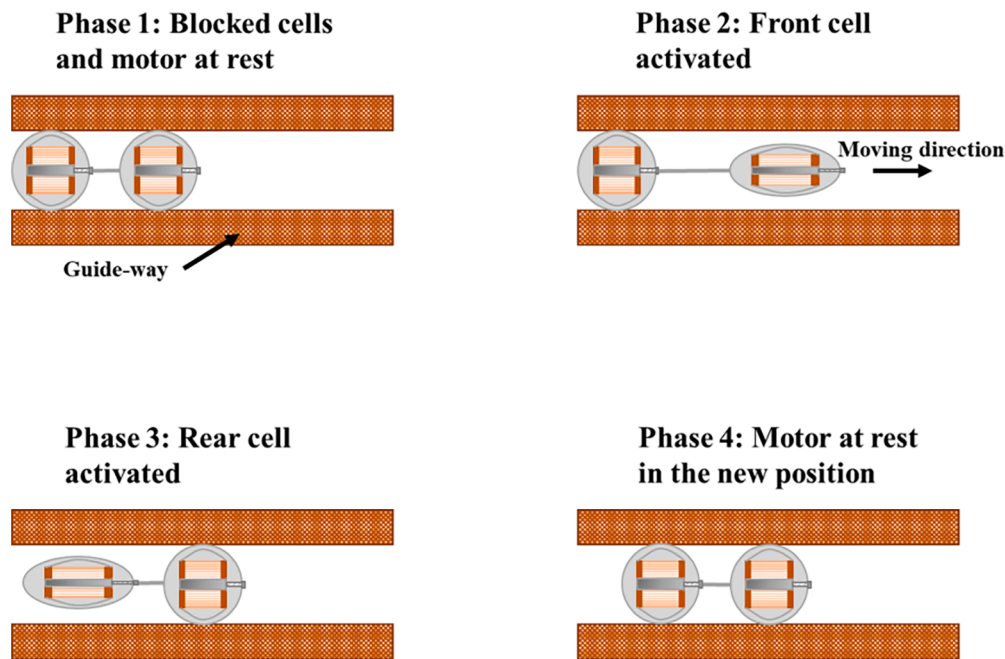


Fig. 16. Sketches of the self-moving linear motor developed in [39].

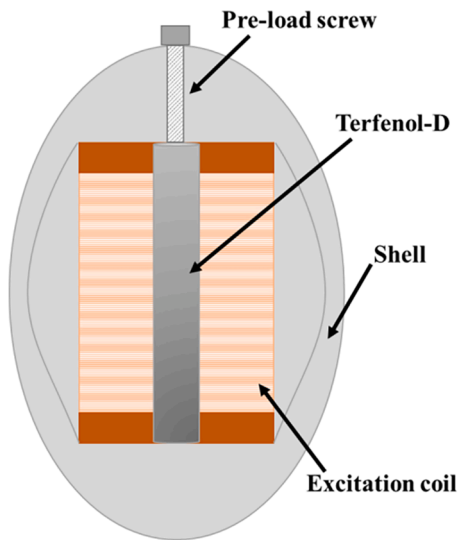


Fig. 17. Schematic of the Terfenol-D actuator and core of the motor developed in [39].

an elliptical vibration in this stator [131]. A model for coupled field iteration (magnetic and mechanical energy coupling and transformation) has been constructed using the finite element approach [42] in order to create the requisite elliptical motion. The Terfenol-D motor was designed to have mechanical resonance with the module's two vibrational modes (a translational mode and a flexure mode) [43]. These two module modes are linked to create elliptical vibrations. To tackle the challenges of inchworm motors, a proof-of-concept magnetostrictive drive rotary motor [41] has been presented and its prototype shown in Fig. 18. Three magnetostrictive actuators are connected to a stator frame, and a rotor is located in the centre of the stator. To integrate actuators and provide pure rotation movement, a flexure hinge mechanism was employed. However, based on the Wiedemann effect [44], a proof of concept for a magnetostrictive motor employing a single Terfenol-D rod has been developed. A magnetic field created by toroidal

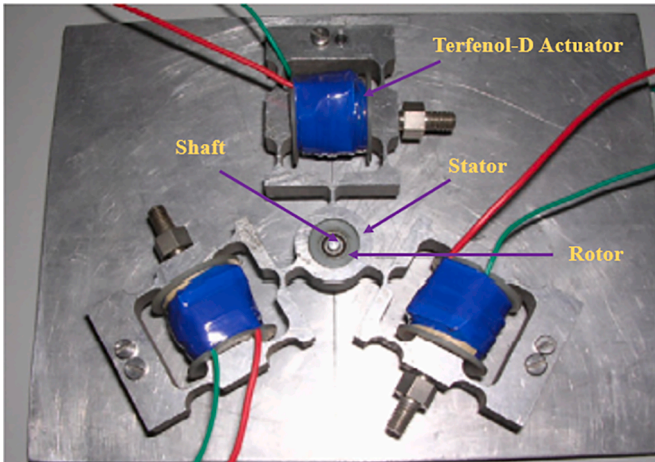


Fig. 18. Prototype of Terfenol-D actuated rotary motor developed in [41].

and solenoid coils was employed to twist Terfenol-D rod in a Sawtooth-like driving signal. The speed and torque parameters of several Terfenol-D motors are shown in Table 2.

Despite the fact that the concept proven to be practicable, the road to commercialization appears to be longer owing to size and expense. Furthermore, it can only provide rotations with one degree of freedom. As a result, a paradigm change is required to develop a new Terfenol-D

Table 2
Terfenol-D motor speed and torque.

Reference	Types of Motor	Speed (rpm)	Torque (N – m)
Ref. [45]	Resonant	0.5	3.0
Ref. [40]	Resonant	0.6	12.2
Ref. [43]	Inchworm	16.7	2.1
Ref. [42]	Inchworm	Not reported	Not reported
Ref. [41]	Inchworm	22 rpm with load & 18 rpm without load	Not reported
Ref. [44]	Resonant	60	6.6

motor with less magnetostrictive material and more degrees of freedom rotations. The innovative MAGMOTOR idea [132] employs a helical magnetic field. Park et al. [44] presented the first proof of concept with a single Terfenol-D rod and helical magnetic field. When a longitudinal magnetic field and a circumferential magnetic field are applied concurrently to a Terfenol-D rod, a helical magnetic field is produced. The circumferential magnetic field is created by sending current through the component or a conductor surrounding by the component. Linear magnetostrictive motors (LMM) [88] have been demonstrated. The powering system that supplies the bias magnetic field, which substitutes the usual hollow cylindrical permanent magnet, is composed of a field coil and a permanent magnet. Instead of a direct current, this LMM field coil was driven by a pulse width modulated voltage with the same frequency and duty cycle as the activation coil. This reduces Joule dissipation.

8.2. Terfenol-D actuated fuel injectors

Integration and acceptance of multifunctional materials are becoming obvious in new engine technologies as essential choices for the development of fuel injection applications at a time when emission reduction is crucial. Fig. 19 shows the prototype of Terfenol-D actuated fuel injector.

Several studies have demonstrated that altering the rate and shape of fuel injection cycles, as well as using several injections per cycle, can greatly improve internal combustion engine fuel economy and reduce NOx and particle emissions. The need for multiple injection capability during power combustion cycle has inspired research into the development of high speed, low cost magnetostrictive and piezoelectric actuator. These actuators may be able to react twice as quickly as solenoid actuators. The engine designer may match the fuel system and engine after treatment needs for best performance because of the engine's inherent quick response time. Overstressed or overheated piezoelectric actuators lose their characteristics. The shortcoming has prompted experts [9–12,15,91,133,134] to look into the issues that arise as a result of the multiple injection capabilities at high pressure.

For the piezoelectric as well as Terfenol-D actuator to operate the ball-valve of a common rail injector, significant displacement and quick response speed are critical criteria. To boost the actuator's response speed, reduce the coil's stimulating time. As a result, raising input voltage slows the growing pace of coil current, allowing the actuator to deliver the needed displacement in a short period of time [14,91]. However, if the high voltage is maintained for an extended period of

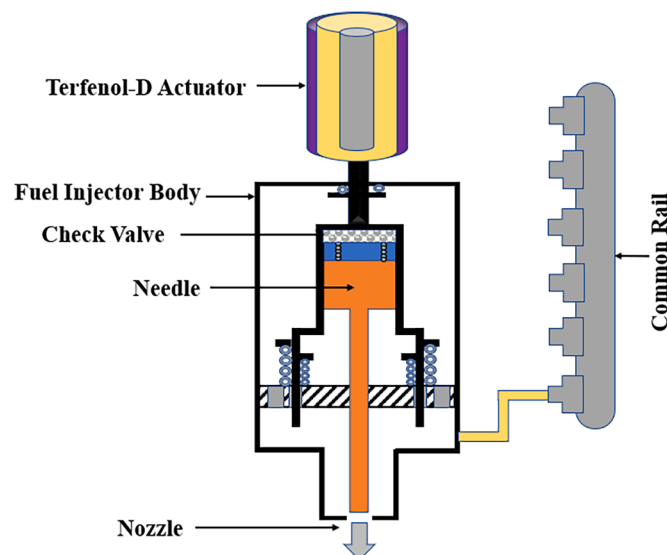


Fig. 19. Proposed prototype of Terfenol-D actuator operated fuel injector.

time, the coil will overheat. As a result, a driving approach may be used to enhance electricity economy while also reducing response time. This voltage waveform has the features of initially applying a high voltage to open the ball valve and then using a low voltage to maintain the displacement, which increases the responsiveness efficiently. Furthermore, because high voltage is maintained for such a short period of time, overheating is prevented. Table 3 shows the performance metrics supplied by the researcher who experimented on Terfenol-D fuel injectors.

The author who uses many technologies and instruments, such as linear magnetostrictive motors to manage the combustion process are the electronically controlled injectors, which enable the exact control of the mass flow rates injected in real-time. The use of the linear magnetostrictive motor to modulate fuel intake on rocket motors is, thus, an example [88].

8.3. Terfenol-D driven inkjet print head

Inkjet printing is demanded for a range of applications because of the technology (Drop-on-demand mode or continuous stream mode) and the versatility in the material (simple ink, conductive ink, different polymers, metal and ceramics). The fields of use have expanded from flexible electronics to organic and 3D printing. Commercially accessible inkjet heads can be classed into three major categories such as thermal, piezo and electrostatic inkjet head.

Motivated researchers might therefore route their research to investigate a magnetostriuctive inkjet printer head [35] which, whenever necessary with lowest droplet volume [135,136] and a change in driving waveform [136], may generate a sound droplet with high viscosity fluids. Fig. 20 shows the magnetostriuctive actuated ink jet printer. The fundamental driven waveform for the inkjet magnetostriuctive head must be a bipolar waveform. However, the W-formed conduction waveform reduces droplet volume and achieves a 30% volume decrease. The Ohnesorge number has a large impact on viscosity and surface tension when ejecting high viscosity fluid with a magnetostriuctive inkjet head. A low Ohnesorge number suggests that flow viscosity is constrained, whereas a high value suggests that surface tension dominates the flow [135]. To produce a droplet, the threshold input current ($\pm 0.3A$) is required. It equates to a force of 23 N from the magnetostriuctive actuator and a displacement of 10 μm . As the input current rises, both droplet speed and volume rise linearly. It will be beneficial to get a faster droplet speed while maintaining a smaller volume. At a holding time of 300 μs , the droplet volume is unsuccessful; however, when the holding time is increased to 360 μs , the droplet volume becomes successful. As a result, by changing the applied waveforms, the magnetostriuctive print head can generate droplets of varying sizes (3.3 nL to 17 nL). Zhou et al. [105] has also recovered the thermal and precision attributes of the magnetostriuctive jet dispenser using a highly accurate driving field and water cooling system.

8.4. Terfenol-D servovalves

Servo valves are normally two-stage equipment with hydraulic amplifiers in the first stage and a second one with several shapes such as a sliding spool, a flapper nozzle, a jet pipe, and a deflector jet. The pressure drop generated by the hydraulic amplifier drives the second stage,

Table 3
Performance parameters of fuel injector.

Researchers	Response Time	Displacement (μm)
Ref. [9]	Not reported	50
Ref. [10]	0.03 seconds	38.3
Ref. [11]	Not reported	Not reported
Ref. [12]	Not reported	Not reported
Ref. [13]	15 milliseconds	50
Ref. [14]	0.92 milliseconds	32
Ref. [134]	0.9–0.5 milliseconds	35

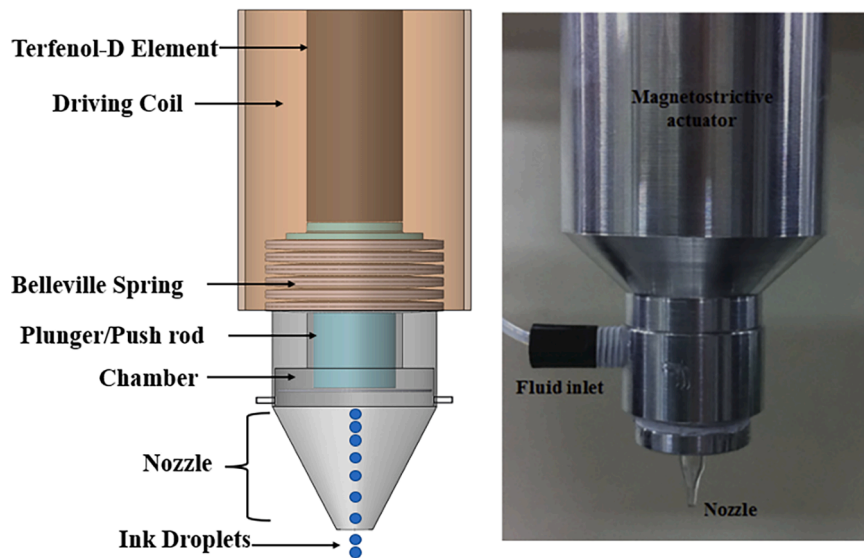


Fig. 20. Proposed prototype of Terfenol-D actuator operated ink jet print head [35].

which regulates valve flow. A hydraulic amplifier is connected to an electromagnetic actuator in electrohydraulic servovalves. The valve can now be modulated by an electrical signal. A fluctuating analog and digital input signal can be converted to a steady hydraulic output (flow or pressure). Servo valves are tiny, precise, high bandwidth valves that are commonly employed in aeromechanical and industrial defence applications requiring high power and quick response.

In two-stage servo valves, the actuation force on the spool can become 10 times larger than the opposed flow forces. The flow forces at the spool have a minor impact on valve dynamics, thanks to this huge force differential. The initial phase flow has a significant impact on the dynamics. The first stage actuator determines the flow parameters. The torque motor has remained the most popular method of actuating servo valves. As a result, numerous researchers [56–58,137,138] have substituted a Terfenol-D actuator for the traditional electromagnetic torque motor.

As shown in Table 4, researchers were driven by a variety of reasons to increase the performance of servo valves. Urai et al. [53–55] constructed a servo valve, commencing with direct drive valve, employing a Terfenol-D actuator between 1993 and 2001. They boosted flow from 2 to 50 l/min with the stroke-expansion system and Pascal's principle. A flapper nozzle servovalve [56] using a Terfenol-D actuator, which is quicker than a normal torque motor, has been constructed and demonstrated for valve step response time (7.8 milliseconds). Zhu et al. [137], Zhu et al. [58] and Zhu et al. [138] also designed a deflector jet servo valve using the Terfenol-D actuator. To assess the tradeoff between input power and rated flow, they increased the number of nozzles from two to four. It has also been observed that three nozzle flapper servo valves outperform standard double nozzle flapper servovalves in terms of performance. The dynamics of a hydraulic displacement amplifier and sliding valve were illustrated experimentally using a multi-coupled valve consisting of direct drive and a Terfenol-D actuator with hydraulic amplifier [113].

8.5. Terfenol-D pump

Smart material pump research has accelerated in the last 10 years, owing to significant advancements in piezoelectric and magnetostrictive materials, advancements in power electronics, and a growing need for high density actuation systems. Over the past many years, the University of Maryland has been developing a number of piezoelectric and magnetostrictive pumps. Gerver [47], who built a Terfenol-D water pump capable of producing 1.83 m³/s of flow rate and a pressure of 5

Table 4

Progress of research in field of Terfenol-D driven servovalves.

Reference	Motivation	Rated Flow ($\frac{l}{min}$)	Step Response (ms)	Frequency bandwidth Hz(–3 dB)
Ref. [53]	Direct drive servo valve	2	<1	650
Ref. [54]	Stroke expansion system	6	1	250
Ref. [55]	Tandem actuator	50	Not reported	250
Ref. [56]	Flexure amplifier & magnetically biased GMA	8	7.8	Not reported
Ref. [137]	For movement of jet nozzle with fast response	7	3	350
Ref. [57]	Direct drive valve with hydraulic amplifier	30	Not reported	110
Ref. [58]	Rapid response and large flow rate with four-flapper nozzle	Not reported	Not reported	Not reported
Ref. [138]	Three nozzle structure with higher flow rate	11.28	2.5	110

psi, was one of the first to publish work on magnetostrictive pumps. Cusak's magnetostrictive pump patent, filed in 1986, has the stated goal of "creating a hydraulic pump capable of creating a pressure based on the magnetostrictive effect." The Terfenol-D pump, capable of produced 1100 psi no-flow pressure and 0.28 in³/s no-load flow rate was created by M. Rupinsky and Dapino. At 10 Hz maximum pressure and 165 Hz maximum flow speed is observed. Fig. 21 shows the structure of Terfenol-D actuator operated pump assembly.

The ability to actuate over 165 Hz was hampered by a high friction hydraulic cylinder. As a result, Nosse and Dapino created a unique magnetorheological rectification valve for the Terfenol-D pump that has a flow rate of 0.25 in³/s and a stroke of 6 inches at 0.5 Hz. The performance of piezoelectric, electrostrictive and Terfenol-D pumps with a driving element of identical size was studied by Larson and his collaborators [48,49]. The electrostrictive (Lead magnesium niobate ceramics (PMN)) and magnetostrictive (Terfenol-D) systems functioned similarly, with blocked pressures of 87 psi and 66 psi and no-load flow rates of 1.6

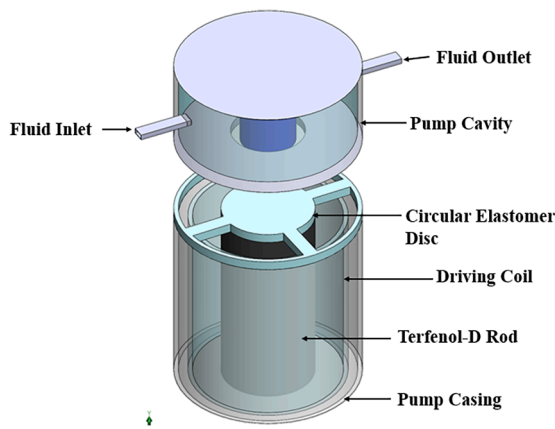


Fig. 21. Structure of Terfenol-D actuator operated pump [48].

in^3/s and $1.48 \text{ in}^3/\text{s}$, respectively. Larson et al. [49] also created a bi-directional actuation system that makes use of a Terfenol-D pump with magnetorheological valves arranged in a Wheat-stone bridge. The bi-directional mechanism can actuate a 1.5 lb load at 3 Hz and a 0.3-inch stroke at 100 Hz. To the best of the author's knowledge, this is the most sophisticated implementation of a bi-directional smart material electrohydraulic actuator that has been published.

Bridger's work [50] has examined the pump for meso-sized magnetostrictive pumps in the highest pumping frequency (2.4 kHz) found in literature to date. With a 325 Hz pumping frequency, Chaudhuri et al. [51] employed two different lengths of Terfenol-D (51 and 112 mm) with a 12.7 mm diameter to achieve 24.8 and $22.7 \text{ cm}^3/\text{s}$ of flow rate. A dynamic model of a magnetostrictive hydraulic pump [46] for frequency, velocity output and load lifting capabilities was constructed as well.

8.6. Active vibration control through Terfenol-D actuators

Active structural vibration control is a technology that uses a time-varying counterforce to bring an oscillating structure to a stop. Previous research has used the Joule magnetostriction to create magnetostrictive actuators that can create dynamic forces to offset external vibrational disturbance. How the Terfenol-D actuators are integrated with the main structure determines the efficiency of active control schemes. Terfenol-D material has a greater counterforce [19,20] and ten times larger strains [21] than piezoelectric materials, therefore it may be used to control vibrations that need forces and displacement amplitudes that piezoelectric actuators cannot supply. Eddy current suppression of Terfenol-D response at high frequencies appears to limit Terfenol-D solid rod applications to kilohertz frequencies, which is substantially less than the megahertz range achievable with piezoelectric materials. Terfenol-D, on the other hand, is still 'fast enough' to be used in vibration control applications, where frequencies are often as low as 10 kHz for structural difficulties and as high as 20 kHz for acoustic radiation concerns.

A novel class of magnetostrictive mini-actuators [22–24] has been designed to reduce undesirable vibrations in composite structures such as aviation and spacecraft components. The tiny magnetostrictive actuator was placed in the cantilever beam to dampen vibrations caused by tip excitations. Because of the existence of two magnetostrictive rods in tandem, the actuator's achievable force was doubled. Because the motion of the accelerometer signal is not totally blocked, the theoretical and empirically measured blocking force of an actuator varies greatly, as does the absence of a precise modulus of elasticity and force sensor components [23]. One of the key results is that while thermal expansion helps to suppress vibrations, if actuation is sustained beyond the crucial time period, the thermal strain causes additional vibrations. Thermal

effects caused by resistive heating of coils have been investigated in order to control vibrations of flexible structures utilising a linear, homogeneous, and isotropic cantilever beam [24] as shown in Fig. 22. In 2012, the Terfenol-D inertial actuator [139] was created for vibrational control, specifically at frequencies less than 2 kHz as shown in Fig. 23. Active vibration control at low frequencies requires a low resonance frequency of the Terfenol-D actuator. For 400 mA currents and a suppression duration of 0.8 sec, the extension bending actuation is less efficient than the pure bending actuation. Instead of using a layer, one can use the Terfenol-D patches in composite beams, plates or shells for the vibration suppression application as shown in Fig. 24.

The Terfenol-D actuator is designed to regulate low-frequency vibrations (1–10 Hz) in civil engineering constructions [16]. Braghin and his colleagues [25,26] employed a Terfenol-D actuator and independent modal control approaches to regulate vibrations produced on the bulkhead of a cruise ship. The Terfenol-D actuator's input current and transmitted force are anticipated using computational and experimental transfer functions at 40 to 2000 Hz. They discovered that the transfer function derived from monoharmonic and sweep sine input stimulation is not significantly different.

Swash plate mechanism, servo flap idea, and higher harmonic control-individual blade control are some of the ideas that may be used to regulate rotor blade vibration. Theoretical and practical work on induced strain actuators, the servo flap idea, individual blade control, and higher harmonic control has been studied and reviewed [18]. They designed and reviewed the benchmark specification and feasibility for greater harmonic control-individual blade control using induced strain actuator, in order to regulate the vibration of the rotor blade using the servo-flap concept. The practicality of direct actuation of an induced strain actuator is made feasible by optimum gain, which results in the least amount of displacement loss and internal energy. The trailing edge of a servo flap is actuated by an induced strain actuator with a hydraulic displacement amplifier, which was constructed and tested experimentally at frequencies ranging from 1 to 30 Hz [27]. The strain displacement of this induced strain actuator was 0.120 mm, and the highest output energy was 0.666 J. The induced strain actuator's amplitude and frequency are improved as a result of lower compressibility loss and counter electromotive force in stacks [18,27].

Terfenol-D struts actuation innovation has been suggested for use with the higher harmonic control system to minimise vibration of individual UH-60A rotor blades [28]. The most demanding system requirement for actuation is terfenol-D actuator mass constraint. Friedmann [29] researched a similar design approach. Both researchers were primarily concerned with the development of a low-mass Terfenol-D actuator with a substantially stronger magnetic field. Active vibration control has been utilised on tall or thin structures in Japan utilising Terfenol-D actuators to increase livability or human comfort in the building's vibration environment induced by high winds and mild but frequent earthquakes [30]. Ohmata et al. [140] suggested a three arm style semi-active vibration control device employing Terfenol-D actuators to decrease intricate vibrations created by earthquakes in complicated multi elastic constructions such as pipeline systems and cranes. To design a controller for compliant structures employing Terfenol-D actuator, Zrostlik et al. [141] employed the z-domain model and adaptive filtering techniques with Kelvin-Voigt damping. Fig. 25 shows the physical component and schematics of Terfenol-D actuated hybrid actuator.

A precise machine (Focused Ion Beam) in semiconductor production has been created with an active six degrees of freedom micro vibration control system [31]. The Terfenol-D actuator and the Stewart Platform vibration isolation technology combined to deliver a 30 dB reduction in vibration [32]. With an average of 30 dB attenuation between 10 Hz and 100 Hz disturbance, the system, which included the actuator, Stewart platform, and adaptive filter algorithm, was able to achieve greater isolation and suppression capabilities [17].

A series of computational and experimental experiments were

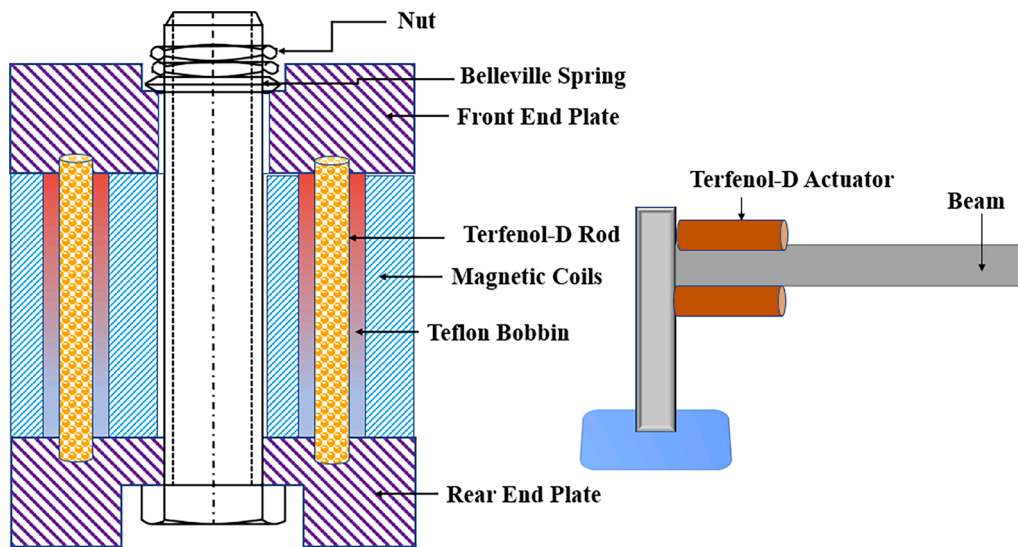


Fig. 22. Schematic diagram of the beam embedded with Terfenol-D actuator [24].

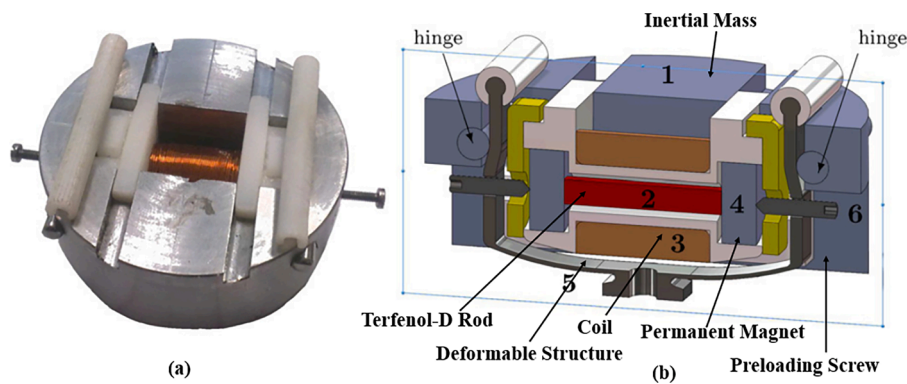


Fig. 23. Low-frequency Terfenol-D inertial actuator (a) prototype (b) section of 3D model [139].

carried out using a basic aluminium beam supported by an actuator at either end and the vibration level lowered by up to 80% [33,34]. The first four modes of a beam were dampened via a model reduction approach and a linear quadratic feedback controller. Experimental energy is supplemented and removed by mounting control from an isolated system to vary the system vibrational noise characteristic [142]. As a control method for vibration control in the turning process, an immune feedback check on the basis of an artificial immune principle is utilised [143]. Not only does the arithmetic for immune feedback control respond to external disturbances but it can also predict that the response goes beyond the stable range. In combination with proportional-integral-derivative (PID) arithmetic, immune arithmetical is a compound controller which can increase the system control performance. As a micro drive for micro vibration removal and accurate positioning, the Terfenol-D actuator have been built and tested [123]. Without an extra vibration isolation mechanism, this Terfenol-D drive can concurrently accomplish large travel and micro vibration removal. The nonlinear constitutive model [144] was developed and used to regulate adaptive vibration using the Terfenol-D actuator. This model incorporated nonlinear constitutive equations, Bouc-Wen equations, Hamilton's principle, the assumed mode method, and the quadric frequency property. The compensating shaker and Terfenol-D actuator test equipment were designed to minimise milling machine tool table vibration [145] by 60-90%. One of the system's downsides is the demand for system self-tuning. Wang and Thomas [146] has created smart tiles for active noise reduction utilising the Terfenol-D actuator for

large-scale underwater applications. For a demanding payload in aerospace applications, the Terfenol-D actuator was utilised to develop and build a passive vibration absorber integrated platform for micro vibration suppression in the low-medium frequency region and submillimeter stroke with nanoscale precision [147,148].

9. Conclusions and outlook

The purpose of this review is to provide a complete assessment of technical advancements in Terfenol-D actuators, notably for motors, valves, injectors, and active vibration control applications. The intention is to provide a comprehensive overview of the development of Terfenol-D actuator technology, including the notion of magnetostriction, magnetic field bias, prestress, and the displacement amplification mechanism. It is envisaged that the development of Terfenol-D actuators would be crucial to construct intelligent structures that can adapt. The hydraulic amplifier is the ideal solution for Terfenol-D actuator displacement amplification. Terfenol-D materials, which combine mechanical and magnetic energy, have enabled novel vibration control methods. Magnetic fields operate the Terfenol-D actuators, making them suited for wireless or non-contact applications. Terfenol-D monolithic materials often have a high electrical conductivity.

As a result, the frequency bandwidth of the Terfenol-D actuators is primarily restricted by eddy currents. Eddy currents are commonly mitigated by lamination or particle embedded composites. Another shortcoming of Terfenol-D actuators is their low deformation. Existing

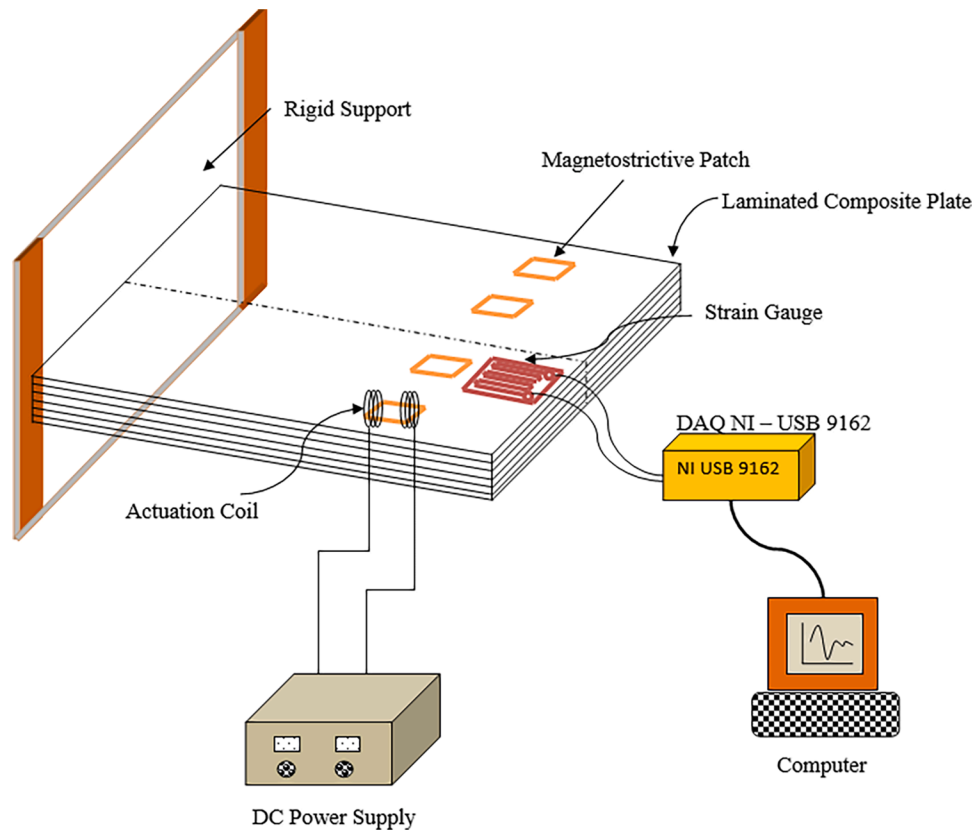


Fig. 24. Active vibration control of laminated composite plate with Terfenol-D patches.

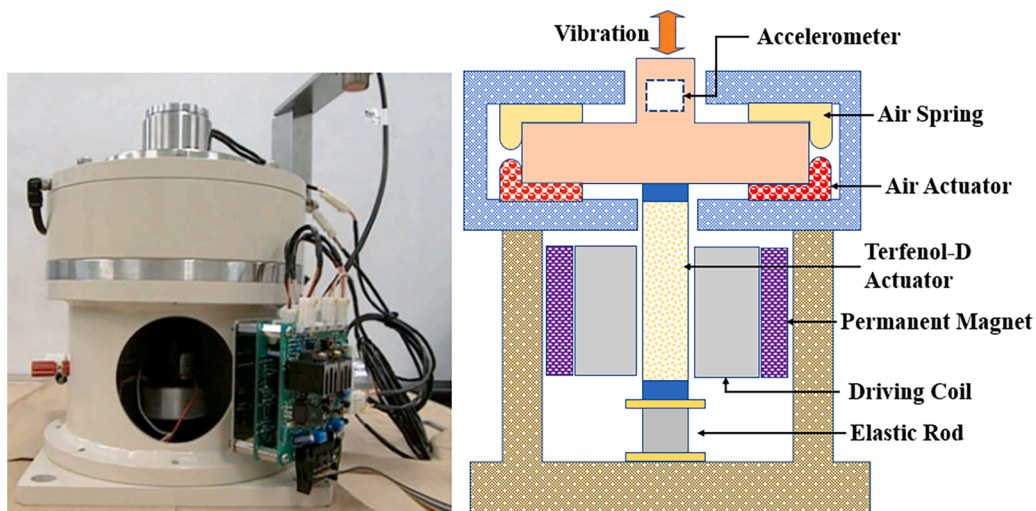


Fig. 25. Physical and its schematic of Terfenol-D actuated hybrid actuator [139].

investigations either create Terfenol-D unimorphs, which convert tiny elongation to enormous bending motion, or develop hybrid actuators, which combine lever mechanisms, flexural mechanisms, hydraulic cylinders, or pneumatic systems to amplify the small deformation. Active vibration control via stiffness or resonance tuning is achieved via the Delta-E effect in Terfenol-D materials. Terfenol-D actuators are still among of the finest alternatives for micro displacements and high forces in situations where cost isn't a problem. The future direction of research is shown in Fig. 26. The following are some subjects which can attract research investigations in the actuator by terfenol-D material and/or magnetostriction.

1. Development and improvement of new flexural-based amplification methods for Terfenol-D actuators with a wider amplification range and greater blocked force delivery. Miniaturization of such amplified Terfenol-D actuators for microelectromechanical and nanoelectromechanical systems applications is also a possibility. This miniaturization opens up a plethora of applications in fields such as precision instrumentation, biomedical devices, and advanced robotics.
2. An essential difficulty to be solved for the structural design of the Terfenol-D fuel injection actuator is the design and development of an effective displacement amplifier to compress the actuator volume

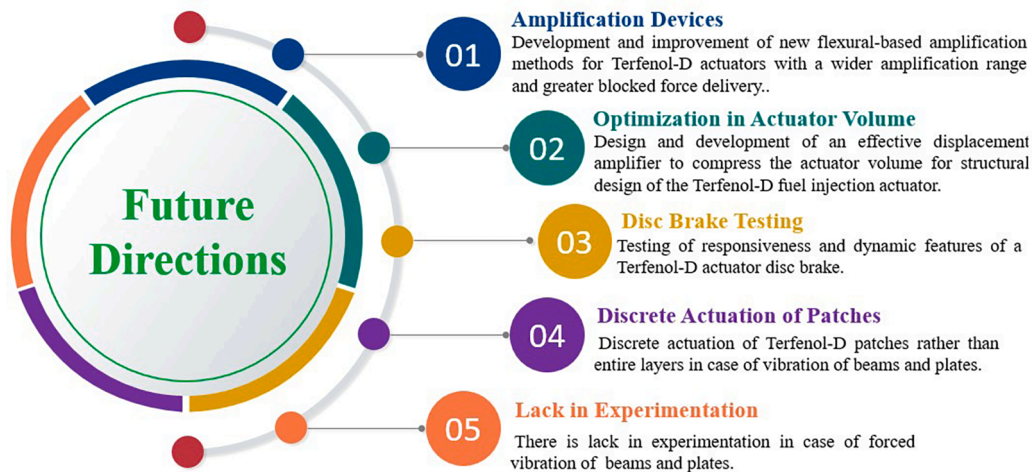


Fig. 26. Future direction of research for Terfenol-D actuator.

while keeping sufficient output force. A power driving circuit with a quick reaction speed, consistent steady-state response, and low sounds should also be considered.

3. Further investigation should prioritize comprehending and enhancing the dynamic functionality of Terfenol-D actuator disc brakes. This involves doing comprehensive experimental investigations to analyze the brake's response to different electrical stimuli and constructing prognostic models. Furthermore, it is essential to optimize the design of the magnetic circuit by conducting analysis and Finite Element Analysis (FEA) and/or COMSOL simulations in order to improve the braking force and efficiency. Additionally, exploring the incorporation of Terfenol-D actuator disc brakes into electric vehicle (EV) braking systems might be a significant area for future study.
4. Researchers should develop and improve discrete Terfenol-D patch vibration control methods for laminated composite beams and plates. Using this strategy gives actuator location and regulating greater flexibility than using full layers. Crucial research issues include: To reduce vibrations most effectively, Terfenol-D patches must be placed and sized optimally. Discrete actuator control algorithm development requires complex control approaches. This method considers feedback, feedforward, and adaptive control. Multi-objective optimization balances design objectives including vibration reduction, energy efficiency, and actuator weight. Discrete Terfenol-D patch actuation is being investigated for aerospace, automotive, and civil engineering applications.
5. A critical knowledge gap exists in the experimental understanding of Terfenol-D layers and patches within laminated composite beams and plates. To address this, future research should focus on experimentally investigating the dynamic behavior of these structures under both free and forced vibration conditions. This includes analyzing the influence of Terfenol-D on structural response to harmonic and random excitations, as well as assessing its effectiveness in attenuating vibrations across a range of frequencies and amplitudes for different practical boundary conditions.
6. The Terfenol-D actuator may be integrated into carbon fiber-reinforced plastic, enabling its use in intelligent structural health monitoring of aircraft structures. This allows for the detection and monitoring of fractures in vital aerospace components such as the airframe and fuselage.

To summarise, this review study aimed to include as many studies as possible, notably in the domain of Terfenol-D actuators. This article is intended to serve as a compilation of research ideas in the subject of Terfenol-D actuators and their usage in diverse intelligent systems

applications.

CRediT authorship contribution statement

Mukund A. Patil: Writing – original draft, Visualization, Resources, Methodology, Data curation, Conceptualization. **Ravikiran Kadoli:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

References

- [1] M.A. Patil, R. Kadoli, Terfenol-D layer in a functionally graded pipe transporting fluid for free vibration. *Pipeline Engineering-Design, Failure, and Management*, IntechOpen, 2023.
- [2] R. Angara, L. Si, M. Anjanappa, A high speed magnetostrictive mirror deflector, *Smart Mater. Struct.* 18 (9) (2009) 095015.
- [3] S. Cao, J. Zheng, B. Wang, W. Huang, L. Wen, B. Wang, A micro-position system of a giant magnetostrictive actuator. *2005 International Conference on Electrical Machines and Systems vol. 3*, IEEE, 2005, pp. 2316–2319.
- [4] X. Chen, C.-Y. Su, High precision control for nano-stage driven by magnetostrictive actuator. *International Conference on Intelligent Robotics and Applications*, Springer, 2013, pp. 666–677.
- [5] H. Nakano, Angstrom positioning system using a giant magnetostriction actuator for high power applications. *Proceedings of the Power Conversion Conference-Osaka 2002 (Cat. No. 02TH8579) vol. 3*, IEEE, 2002, pp. 1102–1107.
- [6] S.F. Tsodikov, V.I. Rakhovsky, Magnetostrictive force actuators for superprecise positioning. *Proceedings ISDEIV. 18th International Symposium on Discharges and Electrical Insulation in Vacuum (Cat. No. 98CH36073) vol. 2*, IEEE, 1998, pp. 713–719.
- [7] B. Yang, G. Meng, Z.-Q. Feng, D. Yang, Giant magnetostrictive clamping mechanism for heavy-load and precise positioning linear inchworm motors, *Mechatronics* 21 (1) (2011) 92–99.
- [8] B.-t. Yang, D.-h. Yang, P.-y. Xu, Y.-d. Cao, Z.-q. Feng, G. Meng, Large stroke and nanometer-resolution giant magnetostrictive assembled actuator for driving segmented mirrors in very large astronomical telescopes, *Sens. Actuators A Phys.* 179 (2012) 193–203.
- [9] H. Chowdhury, S.A. Mazlan, A.G. Olabi, Implementation of magnetostrictive material Terfenol-D in CNG fuel injection actuation. *Advanced Materials Research vol. 47*, Trans Tech Publ, 2008, pp. 630–633.
- [10] L. Li, C. Zhang, B. Kou, X. Li, Design of giant magnetostrictive actuator for fuel injector. *2008 IEEE Vehicle Power and Propulsion Conference*, IEEE, 2008, pp. 1–4.
- [11] L. Pislaru-Dănescu, A.M. Morega, M. Morega, A novel magnetostrictive injection actuator based on new giant magnetostrictive materials. *2011 7TH International*

- Symposium on Advanced Topics in Electrical Engineering (ATEE), IEEE, 2011, pp. 1–6.
- [12] R.G. Yan, Z.J. Wang, L.H. Zhu, Research of the giant magnetostrictive fule injector. *Advanced Materials Research* vol. 889, Trans Tech Publ, 2014, pp. 916–919.
 - [13] M. Ghodsi, N. Hosseinzadeh, A. Özer, H.R. Dizaj, Y. Hojati, N.G. Varzeghani, M. R. Sheykholeslami, S. Talebian, M.H. Ghodsi, A. Al-Yahmadi, Development of gasoline direct injector using giant magnetostrictive materials, *IEEE Trans. Ind. Appl.* 53 (1) (2016) 521–529.
 - [14] G. Xue, P. Zhang, Z. He, D. Li, Y. Huang, W. Xie, Design and experimental study of a novel giant magnetostrictive actuator, *J. Magn. Magn. Mater.* 420 (2016) 185–191.
 - [15] Y. Li, P. Zhang, Z. He, G. Xue, D. Wu, S. Li, Y. Yang, W. Zeng, A simple magnetization model for giant magnetostrictive actuator used on an electronic controlled injector, *J. Magn. Magn. Mater.* 472 (2019) 59–65.
 - [16] P. Bartlett, S. Eaton, J. Gore, W. Metheringham, A. Jenner, High-power, low frequency magnetostrictive actuation for anti-vibration applications, *Sens. Actuators A Phys.* 91 (1–2) (2001) 133–136.
 - [17] T. Zhang, C. Jiang, H. Zhang, H. Xu, Giant magnetostrictive actuators for active vibration control, *Smart Mater. Struct.* 13 (3) (2004) 473.
 - [18] V. Giurgiutiu, Z. Chaudhry, C.A. Rogers, Engineering feasibility of induced strain actuators for rotor blade active vibration control, *J. Intell. Mater. Syst. Struct.* 6 (5) (1995) 583–597.
 - [19] M.B. Moffett, A.E. Clark, M. Wun-Fogle, J. Linberg, J.P. Teter, E.A. McLaughlin, Characterization of Terfenol-D for magnetostrictive transducers, *J. Acoust. Soc. Am.* 89 (3) (1991) 1448–1455.
 - [20] M.B. Moffett, J.M. Powers, A.E. Clark, Comparison of Terfenol-D and PZT-4 power limitations, *J. Acoust. Soc. Am.* 90 (2) (1991) 1184–1185.
 - [21] M.J. Goodfriend, K.M. Shoop, C.G. Miller, High-force high-strain wide-bandwidth linear actuator using the magnetostrictive material Terfenol-D. *Active and Adaptive Optical Components* vol. 1543, International Society for Optics and Photonics, 1992, pp. 301–312.
 - [22] M.A. Anjanappa, J. Bi, Modeling, design, and control of embedded Terfenol-D actuator. *Smart Structures and Materials 1993: Smart Structures and Intelligent Systems* vol. 1917, International Society for Optics and Photonics, 1993, pp. 908–918.
 - [23] M. Anjanappa, J. Bi, A theoretical and experimental study of magnetostrictive mini-actuators, *Smart Mater. Struct.* 3 (2) (1994) 83.
 - [24] M. Anjanappa, J. Bi, Magnetostrictive mini actuators for smart structure applications, *Smart Mater. Struct.* 3 (4) (1994) 383.
 - [25] F. Braghin, S. Cinquemani, F. Resta, Vibrations control in cruise ships using magnetostrictive actuators. *Structural Dynamics*, Volume 3, Springer, 2011, pp. 1207–1214.
 - [26] F. Braghin, S. Cinquemani, F. Resta, A model of magnetostrictive actuators for active vibration control, *Sens. Actuators A Phys.* 165 (2) (2011) 342–350.
 - [27] V. Giurgiutiu, C.A. Rogers, R. Rusovici, Solid-state actuation of rotor blade servo-flap for active vibration control, *J. Intell. Mater. Syst. Struct.* 7 (2) (1996) 192–202.
 - [28] R.C. Fenn, J.R. Downer, D.A. Bushko, V. Gondhalekar, N.D. Ham, Terfenol-D driven flaps for helicopter vibration reduction, *Smart Mater. Struct.* 5 (1) (1996) 49.
 - [29] P. Friedmann, G. Carman, T. Millott, Magnetostrictively actuated control flaps for vibration reduction in helicopter rotors-design considerations for implementation, *Math. Comput. Model.* 33 (10–11) (2001) 1203–1217.
 - [30] T. Fujita, H. Nonaka, C.S. Yang, H. Kondo, Y. Mori, Y. Amasaka, Active vibration control of frame structures with a smart structure using magnetostrictive actuators. *Smart Structures and Materials 1998: Smart Structures and Integrated Systems* vol. 3329, International Society for Optics and Photonics, 1998, pp. 584–595.
 - [31] Y. Nakamura, M. Nakayama, M. Kura, M. Yasuda, T. Fujita, Application of active micro-vibration control system using a giant magnetostrictive actuator, *J. Intell. Mater. Syst. Struct.* 18 (11) (2007) 1137–1148.
 - [32] L.S. Haynes, Z.J. Geng, J.P. Teter, New Terfenol-D actuator design with applications to multiple DOF active vibration control. *Smart Structures and Materials 1993: Smart Structures and Intelligent Systems* vol. 1917, International Society for Optics and Photonics, 1993, pp. 919–928.
 - [33] S.-J. Moon, C.-W. Lim, B.-H. Kim, Y. Park, Structural vibration control using linear magnetostrictive actuators, *J. Sound Vib.* 302 (4–5) (2007) 875–891.
 - [34] S.-J. Moon, C.-W. Lim, B.-H. Kim, Y.-J. Park, Vibration control of a beam using linear magnetostrictive actuators. *Smart Structures and Materials 2005: Damping and Isolation* vol. 5760, International Society for Optics and Photonics, 2005, pp. 341–348.
 - [35] J.-H. Yoo, Y.-W. Park, Development of magnetostrictive inkjet head for liquid droplet formation, *J. Appl. Phys.* 111 (7) (2012) 07A936.
 - [36] A.J. Filipovic, J.W. Sutherland, Assessing the performance of a magnetostrictive-actuated tool holder to achieve axial modulations with application to dry deep hole drilling, *J. Manuf. Process.* 9 (2) (2007) 75–86.
 - [37] W. Panusittikorn, P.I. Ro, Modeling and control of a magnetostrictive tool servo system, *J. Dyn. Syst. Meas. Control* 130 (3) (2008).
 - [38] T. Zhao, H. Yuan, H. Pan, B. Li, Study on the rare-earth giant magnetostrictive actuator based on experimental and theoretical analysis, *J. Magn. Magn. Mater.* 460 (2018) 509–524.
 - [39] J. Kim, J. Doo, Magnetostrictive self-moving cell linear motor, *Mechatronics* 13 (7) (2003) 739–753.
 - [40] J. Vranish, D. Naik, J. Restorff, J. Teter, Magnetostrictive direct drive rotary motor development, *IEEE Trans. Magn.* 27 (6) (1991) 5355–5357.
 - [41] J. Zhou, M. Huang, X. Wang, W. Song, Development of a magnetostrictive drive rotary motor driven by circular parallel movement, *Smart Mater. Struct.* 16 (6) (2007) 2063.
 - [42] Z. Cao, J. Cai, Design of a giant magnetostrictive motor driven by elliptical motion, *Sens. Actuators A Phys.* 118 (2) (2005) 332–337.
 - [43] F. Claeysen, N. Lhermet, R. Le Letty, P. Bouchilloux, Design and construction of a resonant magnetostrictive motor, *IEEE Trans. Magn.* 32 (5) (1996) 4749–4751.
 - [44] J. Park, O. Oh, Y. Park, N. Wereley, A novel concept and proof of magnetostrictive motor, *IEEE Trans. Magn.* 49 (7) (2013) 3379–3382.
 - [45] T. Akuta, An application of giant magnetostrictive material to high power actuators. *10th International Workshop on Rare-earth Magnets and Their Applications*, 1989, 1989.
 - [46] A. Chaudhuri, J.-H. Yoo, N.M. Wereley, Design, test and model of a hybrid magnetostrictive hydraulic actuator, *Smart Mater. Struct.* 18 (8) (2009) 085019.
 - [47] M.J. Gerver, J.H. Goldie, J.R. Swenbeck, R. Shea, P. Jones, R.T. Ilmonen, D. M. Dozor, S. Armstrong, R. Roderick, F.E. Nimblett, et al., Magnetostrictive water pump. *Smart Structures and Materials 1998: Smart Structures and Integrated Systems* vol. 3329, International Society for Optics and Photonics, 1998, pp. 694–705.
 - [48] J.P. Larson, M.J. Dapino, Design of a smart material electro-hydraulic actuator with improved frequency bandwidth. *Industrial and Commercial Applications of Smart Structures Technologies 2012* vol. 8343, International Society for Optics and Photonics, 2012, p. 83430K.
 - [49] J.P. Larson, M.J. Dapino, Reliable, high-frequency miniature valves for smart material electrohydraulic actuators, *J. Intell. Mater. Syst. Struct.* 23 (7) (2012) 805–813.
 - [50] K. Bridger, J.M. Sewell, A.V. Cooke, J.L. Lutian, D. Kohlhafer, G.E. Small, P. M. Kuhn, High-pressure magnetostrictive pump development: a comparison of prototype and modeled performance. *Smart Structures and Materials 2004: Industrial and Commercial Applications of Smart Structures Technologies* vol. 5388, International Society for Optics and Photonics, 2004, pp. 246–257.
 - [51] A. Chaudhuri, J.-H. Yoo, N. Wereley, Dynamic modeling of magnetostrictive hydraulic pump. *ASME 2006 International Mechanical Engineering Congress and Exposition*, American Society of Mechanical Engineers Digital Collection, 2006, pp. 331–337.
 - [52] N. Lhermet, O. Delas, F. Claeysen, Magnetostrictive pump with piezo active valves for more electrical aircraft. *International Conference on New Actuators*, 2006.
 - [53] T. Urai, T. Sugiyama, Development of a direct drive servo valve using a giant magnetostrictive material. *Proceedings of the JFPS International Symposium on Fluid Power* vol. 1993, The Japan Fluid Power System Society, 1993, pp. 131–135.
 - [54] U. Takahiro, Development of a giant magnetostrictive actuator with stroke expansion system and application to a servo valve, *Jpn. Soc. Appl. Electromagn. Mech.* 3 (1995) 46–90.
 - [55] T. Urai, H. Tanaka, Development of a giant magnetostrictive tandem actuator and application to a servovalve, *Nihon Yukuatsu Gakkai Ronbunshu* 32 (3) (2001) 53–57.
 - [56] S. Karunanidhi, M. Singaperumal, Design, analysis and simulation of magnetostrictive actuator and its application to high dynamic servo valve, *Sens. Actuators A Phys.* 157 (2) (2010) 185–197.
 - [57] Z. Yang, Z. He, D. Li, G. Xue, X. Cui, Hydraulic amplifier design and its application to direct drive valve based on magnetostrictive actuator, *Sens. Actuators A Phys.* 216 (2014) 52–63.
 - [58] Y. Zhu, X. Yang, X. Wang, Development of a four-nozzle flapper servovalve driven by a giant magnetostrictive actuator, *Proc. Inst. Mech. Eng. Part I J. Syst. Control Eng.* 229 (4) (2015) 293–307.
 - [59] A.Q. Xu, Study on the dynamic characteristics of a high frequency brake based on giant magnetostrictive material, *Smart Mater. Struct.* 25 (6) (2016) 065001.
 - [60] S. Valadkhan, K. Morris, A. Khajepour, Review and comparison of hysteresis models for magnetostrictive materials, *J. Intell. Mater. Syst. Struct.* 20 (2) (2009) 131–142.
 - [61] Z. Deng, M.J. Dapino, Review of magnetostrictive vibration energy harvesters, *Smart Mater. Struct.* 26 (10) (2017) 103001.
 - [62] G. Xue, P. Zhang, X. Li, Z. He, H. Wang, Y. Li, R. Ce, W. Zeng, B. Li, A review of giant magnetostrictive injector (GMI), *Sens. Actuators A Phys.* 273 (2018) 159–181.
 - [63] Z. Deng, M.J. Dapino, Review of magnetostrictive materials for structural vibration control, *Smart Mater. Struct.* 27 (11) (2018) 113001.
 - [64] V. Apicella, C.S. Clemente, D. Davino, D. Leone, C. Visone, Review of modeling and control of magnetostrictive actuators. *Actuators* vol. 8, Multidisciplinary Digital Publishing Institute, 2019, p. 45.
 - [65] J. Atulasimha, A.B. Flatau, A review of magnetostrictive iron–gallium alloys, *Smart Mater. Struct.* 20 (4) (2011) 043001.
 - [66] A. Gray, On the alterations in the dimensions of the magnetic metals by the act of magnetisation, *Nature* 26 (678) (1882) 625–626.
 - [67] A.E. Clark, Magnetostrictive rare earth-Fe₂ compounds, *Handb. Ferromagn.* 1 (1980) 531–589.
 - [68] A.t. Clark, H. Belson, Giant room-temperature magnetostrictions in TbFe₂ and DyFe₂, *Phys. Rev. B* 5 (9) (1972) 3642.
 - [69] Z. Nie, S. Yang, Y.-d. Wang, Z. Wang, D. Liu, Y. Ren, T. Chang, R. Zhang, In-situ studies of low-field large magnetostriction in Tb_{1-x} Dy_xFe₂ compounds by synchrotron-based high-energy X-ray diffraction, *J. Alloys Compd.* 658 (2016) 372–376.

- [70] N. Wang, Y. Liu, H. Zhang, X. Chen, Y. Li, Effect of Nb on magnetic and mechanical properties of TbDyFe alloys, *J. Magn. Magn. Mater.* 449 (2018) 223–227.
- [71] V. Hari Babu, G. Markandeyulu, A. Subrahmanyam, Giant magnetostriction in $Sm_{1-x}Nd_xFe_{1.93}$ compounds, *Appl. Phys. Lett.* 90 (25) (2007) 252513.
- [72] H. ZHANG, P. Yanrong, W. Lijuan, C. JIANG, Magnetostriction and hysteresis of $<110>$ oriented TbDyHoFe_{1.95} alloy, *J. Rare Earths* 28 (2010) 403–405.
- [73] C. Jiang, T. Ma, H. Xu, A kind of wide operating temperature range giant magnetostrictive alloys, *J. Alloys Compd.* 449 (1–2) (2008) 156–160.
- [74] J.A. Chelvane, M. Palit, H. Basumatary, S. Pandian, V. Chandrasekaran, Effects of Ti addition on the microstructure and magnetic properties of magnetostrictive Tb–Dy–Fe alloys, *Scr. Mater.* 61 (5) (2009) 548–551.
- [75] M. Palit, J.A. Chelvane, H. Basumatary, S. Pandian, V. Chandrasekaran, The effect of Nb and Zr addition on the microstructural features and magnetic properties of Tb_{0.3}Dy_{0.7}Fe_{1.95}, *Scr. Mater.* 60 (1) (2009) 56–59.
- [76] C. Jiang, Y. Zhao, L. Xu, H. Xu, Orientation, morphology and magnetostriction of a heat-treated $<110>$ oriented TbDyFe alloy, *J. Alloys Compd.* 373 (1–2) (2004) 167–170.
- [77] W. Wu, M. Zhang, X. Gao, S. Zhou, Effect of two-steps heat treatment on the mechanical properties and magnetostriction of $<110>$ oriented TbDyFe giant magnetostrictive material, *J. Alloys Compd.* 416 (1–2) (2006) 256–260.
- [78] W. Mei, T. Umeda, S. Zhou, R. Wang, Magnetostriction of grain-aligned Tb_{0.3}Dy_{0.7}Fe_{1.95}, *J. Alloys Compd.* 224 (1) (1995) 76–80.
- [79] L. Zheng, W. Li, B. Cui, G.C. Hadjipanayis, Tb_{0.3}Dy_{0.7}Fe_{1.92} nanoflakes prepared by surfactant-assisted high energy ball milling, *J. Alloys Compd.* 509 (19) (2011) 5773–5776.
- [80] H. Uchida, M. Wada, K. Koike, H. Uchida, V. Koeninger, Y. Matsumura, H. Kaneko, T. Kurino, Giant magnetostrictive materials: thin film formation and application to magnetic surface acoustic wave devices, *J. Alloys Compd.* 211 (1994) 576–580.
- [81] J.D. Snodgrass, O. McMasters, Optimized Terfenol-D manufacturing processes, *J. Alloys Compd.* 258 (1–2) (1997) 24–29.
- [82] L. Wu, W. Zhan, X. Chen, Microsegregation phenomenon in Terfenol-D rods grown by electron beam zoning method, *J. Alloys Compd.* 255 (1–2) (1997) 262–265.
- [83] C. Ji, W. Ma, J. Li, Y. Zhou, Technique of melting and casting of (TbDy)Fe₂ matrix alloy bars, *J. Alloys Compd.* 327 (1–2) (2001) 220–223.
- [84] S. Kano, Y. Matsumura, H. Uchida, TbFe₂ giant magnetostrictive alloy prepared by drop-tube system, *J. Alloys Compd.* 408 (2006) 331–334.
- [85] A. Grunwald, A.-G. Olabi, Design of a magnetostrictive (ms) actuator, *Sens. Actuators A Phys.* 144 (1) (2008) 161–175.
- [86] G. Engdahl, I.D. Mayergoyz, *Handbook of Giant Magnetostrictive Materials* vol. 107, Elsevier, 2000.
- [87] A.-G. Olabi, A. Grunwald, Design and application of magnetostrictive materials, *Mater. Des.* 29 (2) (2008) 469–483.
- [88] L. Pislaru-Dănescu, A.M. Morega, M. Morega, F. Bunea, P. Marius, C.A. Băbuțanu, A new type of linear magnetostrictive motor, *Electr. Eng.* 99 (2) (2017) 601–613.
- [89] W. Lei, T. Jiu-bin, Z. Shan, A giant magnetostrictive actuator based on use of permanent magnet, *Int. J. Adv. Manuf. Technol.* 46 (9–12) (2010) 893–897.
- [90] G. Xue, Z. He, D. Li, Z. Yang, Z. Zhao, Analysis of the giant magnetostrictive actuator with strong bias magnetic field, *J. Magn. Magn. Mater.* 394 (2015) 416–421.
- [91] G. Xue, P. Zhang, Z. He, D. Li, Z. Yang, Z. Zhao, Displacement model and driving voltage optimization for a giant magnetostrictive actuator used on a high-pressure common-rail injector, *Mater. Des.* 95 (2016) 501–509.
- [92] R. Joshi, S.M. Rao, R. Kadoli, Finite element magnetostatic analysis of magnetostrictive (Tb_{0.3}Dy_{0.7}Fe_{1.95}) actuator with different housing materials, *Defence Sci. J.* 63 (4) (2013) 423.
- [93] R. Joshi, R. Kadoli, Magneto-mechanical coupled magnetostriction model for Terfenol-D actuator under step input, *Procedia Mater. Sci.* 5 (2014) 154–163.
- [94] K. Kondo, Dynamic behaviour of Terfenol-D, *J. Alloys Compd.* 258 (1–2) (1997) 56–60.
- [95] F.T. Calkins, M.J. Dapino, A.B. Flatau, Effect of prestress on the dynamic performance of a Terfenol-D transducer, *Smart Structures and Materials 1997: Smart Structures and Integrated Systems* vol. 3041, International Society for Optics and Photonics, 1997, pp. 293–304.
- [96] X. Liu, H. Zhang, X. Gao, M. Chen, Design and simulation analysis of giant magnetostrictive actuator, *Mater. Technol.* 30 (3) (2015) 155–158.
- [97] Z. Jia, H. Liu, F. Wang, Researches on the computer digital control system of giant magnetostrictive actuator, *Mechatronics* 19 (7) (2009) 1191–1196.
- [98] H. Zhang, T. Zhang, C. Jiang, Magnetostrictive actuators with large displacement and fast response, *Smart Mater. Struct.* 21 (5) (2012) 055014.
- [99] H. Zhang, T. Zhang, C. Jiang, Design of a uniform bias magnetic field for giant magnetostrictive actuators applying triple-ring magnets, *Smart Mater. Struct.* 22 (11) (2013) 115009.
- [100] Y. Sato, K. Shinohara, Power-saving drive in 2-position control of giant-magnetostrictive actuator, *IEEE Trans. Magn.* 45 (10) (2009) 4554–4557.
- [101] L. Liyi, Z. Chengming, Y. Baiping, L. Xiaopeng, Research of a giant magnetostrictive valve with internal cooling structure, *IEEE Trans. Magn.* 47 (10) (2011) 2897–2900.
- [102] G. Engdahl, Design procedures for optimal use of giant magnetostrictive materials in magnetostrictive actuator applications, *Parameters Parameters* 1 (2002) 2.
- [103] G.P. Carman, M. Mitrovic, Nonlinear constitutive relations for magnetostrictive materials with applications to 1-D problems, *J. Intell. Mater. Syst. Struct.* 6 (5) (1995) 673–683.
- [104] D.C. Meeker, D.M. Dozor, Thermomagnetic optimization of solenoidal magnetostrictive actuators, *Smart Structures and Materials 1999: Mathematics and Control in Smart Structures* vol. 3667, International Society for Optics and Photonics, 1999, pp. 122–131.
- [105] C. Zhou, J.-a. Duan, G. Deng, J. Li, Improved thermal characteristics of a novel magnetostrictive jet dispenser using water-cooling approach, *Appl. Therm. Eng.* 112 (2017) 1–6.
- [106] T. Zhifeng, L. Fuzai, L. Yang, Magnetic field distribution in the cross section of Terfenol-D rod and its application, *J. Rare Earths* 27 (3) (2009) 525–528.
- [107] M. Niu, B. Yang, Y. Yang, G. Meng, Dynamic modelling of magnetostrictive actuator with fully coupled magneto-mechanical effects and various eddy-current losses, *Sens. Actuators A Phys.* 258 (2017) 163–173.
- [108] M. Stuebner, J. Atulasimha, R.C. Smith, Quantification of hysteresis and nonlinear effects on the frequency response of ferroelectric and ferromagnetic materials, *Smart Mater. Struct.* 18 (10) (2009) 104019.
- [109] Y.-W. Park, D.-Y. Kim, Development of a magnetostrictive microactuator, *J. Magn. Magn. Mater.* 272 (2004) E1765–E1766.
- [110] X. Yang, Y. Zhu, Y. Zhu, Characteristic investigations on magnetic field and fluid field of a giant magnetostrictive material-based electro-hydrostatic actuator, *Proc. Inst. Mech. Eng. Part G J. Aerosp. Eng.* 232 (5) (2018) 847–860.
- [111] C. Rong, Z. He, D. Li, Z. Yang, G. Xue, Dynamic modeling and analysis of stack giant magnetostrictive actuator, *Sens. Actuators A Phys.* 276 (2018) 205–218.
- [112] Q. Lu, Q. Nie, X. Jiang, Q. Cao, D. Chen, Magnetostrictive actuator with differential displacement amplification mechanism, *Mechanics* 22 (4) (2016) 273–278.
- [113] Z. Yang, Z. He, D. Li, J. Yu, X. Cui, Z. Zhao, Direct drive servo valve based on magnetostrictive actuator: multi-coupled modeling and its compound control strategy, *Sens. Actuators A Phys.* 235 (2015) 119–130.
- [114] Z. Yang, Z. He, D. Li, X. Cui, G. Xue, Z. Zhao, Dynamic analysis and application of a novel hydraulic displacement amplifier based on flexible pistons for micro stage actuator, *Sens. Actuators A Phys.* 236 (2015) 228–246.
- [115] F. Zeng, J.J. Lou, S.J. Zhu, Study on giant magnetostrictive material with design and simulation of displacement magnifying mechanism, *Advanced Materials Research* vol. 675, Trans Tech Publ, 2013, pp. 219–226.
- [116] S. Chakrabarti, M.J. Dapino, Hydraulically amplified magnetostrictive actuator for active engine mounts, *ASME 2008 Conference on Smart Materials, Adaptive Structures and Intelligent Systems*, American Society of Mechanical Engineers Digital Collection, 2008, pp. 795–802.
- [117] H. Tang, Y. Li, X. Xiao, Development and assessment of a novel hydraulic displacement amplifier for piezo-actuated large stroke precision positioning, 2013 IEEE International Conference on Robotics and Automation, IEEE, 2013, pp. 1409–1414.
- [118] S. Chakrabarti, M.J. Dapino, A dynamic model for a displacement amplified magnetostrictive driver for active mounts, *Smart Mater. Struct.* 19 (5) (2010) 055009.
- [119] V. Giurgiutiu, Z.A. Chaudhry, C.A. Rogers, Stiffness issues in the design of ISA displacement amplification devices: case study of a hydraulic displacement amplifier, *Smart Structures and Materials 1995: Smart Structures and Integrated Systems* vol. 2443, International Society for Optics and Photonics, 1995, pp. 105–119.
- [120] Z. Chaudhry, Design of displacement-amplified induced-strain actuators for maximum energy output, *J. Mech. Des.* 119 (1997) 511.
- [121] D.T. Nosse, M.J. Dapino, Compact actuation through magnetorheological flow control and rectification of magnetostrictive vibrations, *Smart Structures and Materials 2005: Smart Structures and Integrated Systems* vol. 5764, International Society for Optics and Photonics, 2005, pp. 262–273.
- [122] J. Garcia-Bonito, M. Brennan, S. Elliott, A. David, R. Pinnington, A novel high-displacement piezoelectric actuator for active vibration control, *Smart Mater. Struct.* 7 (1) (1998) 31.
- [123] X. Sun, B. Yang, A new methodology for developing flexure-hinged displacement amplifiers with micro-vibration suppression for a giant magnetostrictive micro drive system, *Sens. Actuators A Phys.* 263 (2017) 30–43.
- [124] M. Niu, B. Yang, Y. Yang, G. Meng, Modeling and optimization of magnetostrictive actuator amplified by compliant mechanism, *Smart Mater. Struct.* 26 (9) (2017) 095029.
- [125] Y. Zhu, X. Yang, N.M. Wereley, Theoretical and experimental investigations of a magnetostrictive electro-hydrostatic actuator, *Smart Mater. Struct.* 27 (10) (2018) 105043.
- [126] Z. Wang, Y. Zhu, R. Li, C. Liu, N. Bruno, Development of a dual magnetostrictive material rods-based electro-hydrostatic actuator, *J. Intell. Mater. Syst. Struct.* 30 (13) (2019) 1871–1881.
- [127] Y. Zhu, Z. Wang, S. Fei, Development of a magnetostrictive material-based electro-hydrostatic actuator with an active rotary rectifying valve, *Proc. Inst. Mech. Eng. Part I J. Syst. Control Eng.* 233 (8) (2019) 994–1008.
- [128] J. Zhou, Z. He, C. Rong, G. Xue, A giant magnetostrictive rotary actuator: design, analysis and experimentation, *Sens. Actuators A Phys.* 287 (2019) 150–157.
- [129] J. Zhou, Z. He, Z. Shi, J. Song, Q. Li, Design and experimental performance of an inertial giant magnetostrictive linear actuator, *Sens. Actuators A Phys.* 301 (2020) 111771.
- [130] M. Afzal, L. Kari, I. Lopez Arteaga, Adaptive control of normal load at the friction interface of bladed disks using giant magnetostrictive material, *J. Intell. Mater. Syst. Struct.* 31 (8) (2020) 1111–1125.
- [131] F. Claeysse, N. Lhermet, R. Le Letty, P. Bouchilloux, Actuators, transducers and motors based on giant magnetostrictive materials, *J. Alloys Compd.* 258 (1–2) (1997) 61–73.

- [132] Y.-W. Park, H.-G. Lee, M.D. Noh, Rotary magnetostrictive motor using helical magnetic field, *Int. J. Precis. Eng. Manuf.* 17 (5) (2016) 659–663.
- [133] Z.B. He, G.M. Xue, D.W. Li, Valve opening of the injector controlled by giant magnetostrictive actuator with posterior returning spring, *Applied Mechanics and Materials* vol. 427, Trans Tech Publ, 2013, pp. 174–178.
- [134] G. Xue, P. Zhang, Z. He, B. Li, C. Rong, Design and model for the giant magnetostrictive actuator used on an electronic controlled injector, *Smart Mater. Struct.* 26 (5) (2017) 05LT02.
- [135] Y.-W. Park, O.-K. Oh, M.D. Noh, Ejection feasibility of high viscosity fluid with magnetostrictive inkjet printhead, *Int. J. Precis. Eng. Manuf.* 16 (7) (2015) 1369–1374.
- [136] J.-H. Yoo, Y.-W. Park, Experimental investigation of magnetostrictive DoD inkjet head for droplet formation, *Curr. Appl. Phys.* 11 (1) (2011) S353–S359.
- [137] Y. Zhu, Y. Li, Development of a deflector-jet electrohydraulic servovalve using a giant magnetostrictive material, *Smart Mater. Struct.* 23 (11) (2014) 115001.
- [138] Y. Zhu, X. Yang, T. Fu, Dynamic modeling and experimental investigations of a magnetostrictive nozzle-flapper servovalve pilot stage, *Proc. Inst. Mech. Eng. Part I J. Syst. Control Eng.* 230 (3) (2016) 244–254.
- [139] F. Braghin, S. Cinquemani, F. Resta, A low frequency magnetostrictive inertial actuator for vibration control, *Sens. Actuators A Phys.* 180 (2012) 67–74.
- [140] K. Ohmata, M. Zaike, T. Koh, A three-link arm type vibration control device using magnetostrictive actuators, *J. Alloys Compd.* 258 (1–2) (1997) 74–78.
- [141] R.L. Zrostlik, A.B. Flatau, R.C. Smith, Active vibration control of a compliant structure using a magnetostrictive actuator, *Smart Structures and Materials* 1999: Smart Structures and Integrated Systems vol. 3668, International Society for Optics and Photonics, 1999, pp. 384–394.
- [142] M. Hiller, M.D. Bryant, J. Umegaki, Attenuation and transformation of vibration through active control of magnetostrictive Terfenol, *J. Sound Vib.* 134 (3) (1989) 507–519.
- [143] G. Li, J. Lin, X. Pan, Y. Wei, Research on the turning vibration control system based on giant magnetostrictive actuator, 2007 International Conference on Mechatronics and Automation, IEEE, 2007, pp. 2559–2563.
- [144] T. Zhang, B.T. Yang, H.G. Li, G. Meng, Dynamic modeling and adaptive vibration control study for giant magnetostrictive actuators, *Sens. Actuators A Phys.* 190 (2013) 96–105.
- [145] M. Zucca, F.A. Raffa, A. Fasana, N. Colella, A simplified vibration compensation through magnetostrictive actuators, *J. Vib. Control* 21 (14) (2015) 2903–2912.
- [146] W. Wang, P. Thomas, Low-frequency active noise control of an underwater large-scale structure with distributed giant magnetostrictive actuators, *Sens. Actuators A Phys.* 263 (2017) 113–121.
- [147] X. Wang, H. Wu, B. Yang, Micro-vibration suppressing using electromagnetic absorber and magnetostrictive isolator combined platform, *Mech. Syst. Signal Process.* 139 (2020) 106606.
- [148] X. Sun, Y. Liu, W. Hu, T. Sun, J. Hu, Design optimization of a giant magnetostrictive driving system for large stroke application considering vibration suppression in working process, *Mech. Syst. Signal Process.* 138 (2020) 106560.